Geometric and Learned Relay Control for Connectivity-Constrained Drone Delivery in Disaster Response

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Abstract

In disaster relief, delivery drones must stay connected to a depot through a multi-hop relay chain formed by the fleet itself, while carrying supplies with limited capacity and returning to reload. We model this as a makespan-minimisation problem in which every drone can either deliver or relay, and connectivity is a hard constraint enforced at every time step. We propose two controllers that share the same connectivity-preserving projection: a geometric chain-relay rule with a deadlock-recovery mechanism, and a fleet-size-agnostic policy learned once with discrete soft actor-critic over a set-based, autoregressive manager. Both are compared on identical instances with a paired protocol across fleets of 3 to 12 drones and 20 to 400 demand points, against four learning-free baselines, a rollout-lookahead version of the rule, and a time-expanded reference model solved exactly on small instances. The learned policy completes every instance in every configuration; the rule completes 98–100% with the recovery mechanism and 62% without it. The two controllers divide the configuration space: with three drones the rule is 30–39 steps faster in paired median, with four drones the two are statistically indistinguishable over 480 paired runs, and with five or more drones the learned policy is faster by 38 to 178 steps. The policy is trained only on three and four drones and transfers without retraining to every larger fleet tested; the set-based architecture carries completion and training on randomised configurations carries speed. The three-drone deficit is localised to a single decision, which drone relays, and eleven policy-side interventions covering information, credit assignment, decision structure, initialisation and reward shaping all fail to remove it; half of it disappears when the connectivity projection is changed from radial shrinking to tangential sliding, showing that the deficit is partly a property of how the constraint is enforced rather than of the learner. On fifty instances solved to proven optimality, including ten that force reloading, and run in the model's own decision space, the rule and the learned policy are about 55% above the optimum; a one-step rollout lookahead on the rule, which evaluates each candidate assignment by simulating the rule to completion, brings the rule to 14% at tens of seconds per instance, offline searches over visit orders and over task sequences are 23% and 6% above it, and a beam search in the model's state space is 0.4% above it, returning the proven optimum on 46 of 50 instances. The same lookahead applied to the learned policy brings it to 19%, whereas fine-tuning it on the small instances removes its tail but not its median gap. In the continuous space the lookahead rule is also 13–25% faster than both online controllers on every configuration tested, at 1–6 minutes per instance against milliseconds per decision, so the learned policy's advantage is its latency and its scaling rather than its proximity to the optimum.

Keywords: drone delivery, relay network, connectivity constraint, multi-agent reinforcement learning, rule, disaster logistics

1. Introduction

When communication, logistics and road infrastructure are destroyed, a fleet of drones can carry emergency supplies from a temporary depot to many demand points. Drones have been studied for last-mile distribution in disaster relief [1, 2, 3] and for medical delivery [4, 5], and the routing of drone fleets has grown into a literature of its own since the flying-sidekick travelling salesman problem [6, 7, 8, 9]. Two features make the disaster setting different from parcel delivery. First, every drone must be reachable by the depot at all times, because loading and unloading are supervised remotely and the delivery status must be tracked; with the ground network gone, reachability is provided only by the drones themselves acting as radio relays in a multi-hop chain. Second, the fleet is small and each drone carries a few units, so drones shuttle between the depot and the demand points and every drone that stands still as a relay is a drone that is not delivering. The operational question is how to assign delivery and relay roles over time so that the last demand point is served as early as possible without ever breaking the chain. Aerial relays are well studied in the communications literature [10, 11, 12], and connectivity maintenance is a classical topic in multi-robot control [13, 14], but in both the relay *is* the mission; here relaying competes with the delivery mission for the same vehicles.

Three properties make the problem hard. A hard connectivity constraint creates deadlocks: a greedy assignment that sends every drone to its nearest demand point completes none of the standard instances in our study, because the fleet pulls itself apart until no drone can move. Relaying competes directly with throughput, so the right number and identity of relays changes

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with the geometry of the remaining demand points and with how many drones are left with cargo. And exact optimisation does not scale: a space–time network model [15, 16] on a 98-metre lattice is solved to proven optimality in seconds while no reloading is needed, but once drones must reload the mixed-integer solver proves optimality on few instances within an hour and returns no feasible solution on some, whereas drone-routing models without a connectivity constraint are solved exactly for tens of customers [17].

Proposal. We propose three methods for this problem and study them side by side: two online controllers that decide in milliseconds, and one offline search that we use as a quality reference.

1. A *geometric chain-relay rule*. The drone farthest from the depot with cargo is the lead; the number of relays is the number of communication hops its target needs; the relays are the next-farthest drones, stationed on the segment from the depot to the lead at intervals of $0.9R$. Because the rule is deterministic it can repeat a deadlocked assignment indefinitely, so we add a *deadlock-recovery* mechanism that swaps the lead whenever the fleet has stalled for a fixed number of steps. The recovery mechanism alone lifts completion on the standard configuration from 62% to 98%.
2. Two *search heuristics*, evaluated by simulation and offline. The first keeps the rule's relay geometry and searches only the per-drone visit order; the second also searches the relays, that is, where each drone stands as a relay and until which delivery, as a sequence of tasks per drone. They cost minutes to an hour per instance, and we use them to separate how much of the distance to the optimum is the choice of targets, how much is the relay geometry, and how much is left.
3. A *learned policy* that is agnostic to fleet size. A manager assigns each drone one of seven discrete options every twenty steps; it encodes the drones, candidate demand points and relay slot as a set, pools them with attention, and decides autoregressively so that a formation of one lead and k relays can be composed deliberately. It is trained once with discrete soft actor-critic [18] on fleets of three and four drones and then run without retraining on fleets of up to twelve, in the spirit of size-agnostic learned solvers for routing [19, 20].

All three act through the same connectivity-preserving projection: each commanded move is scaled by the largest factor that keeps the fleet connected. Connectivity is therefore guaranteed by construction for both, and the empirical question is whether a controller makes progress under the constraint rather than whether it violates it.

Research questions. We organise the study around five questions.

RQ1 Do the proposed controllers always complete the delivery while connectivity is enforced at every step, and when they do not, why not?

RQ2 On small instances, how close are the proposed controllers to the optimal solution of a space–time network model solved exactly?

RQ3 How does the performance of the two controllers compare across operating conditions, in particular across fleet size, number of demand points, depot position and communication radius, and where do the differences come from?

RQ4 Does the learned policy, trained on fleets of three and four drones, transfer upward to larger fleets and larger instances without retraining, and what carries the transfer?

RQ5 Are the components of each controller necessary?

Findings. The learned policy completes every instance in every configuration tested; the rule completes 98–100% with the recovery mechanism (RQ1). On forty instances solved to proven optimality and run in the model's own decision space, the rule and the learned policy are about 55% above the optimum, an offline search over visit orders is 23% above it and a search over task sequences that also decides the relays is 6% above it, so the distance is first the myopic choice of targets and then the parked relays; a one-step rollout lookahead on the rule closes the first part (14% above the optimum), a beam search in the model's state space reaches the optimum on 46 of 50 instances (0.4%), and the lookahead applied to the learned policy brings it to 19%, while fine-tuning the policy on the small instances removes its tail but not its median (RQ2). The two controllers divide the configuration space by fleet size: the rule is faster with three drones, the two are statistically indistinguishable with four, and the learned policy is faster with five or more, by 62 to 178 steps up to ten drones and 76 at twelve; the three-drone deficit is a single decision, which drone relays, and it resists eleven policy-side interventions but shrinks by half when the connectivity projection is changed (RQ3). The learned policy never saw more than four drones or 80 demand points in training, yet it completes every instance with up to twelve drones and 400 demand points; the set-based architecture carries completion and the configuration randomisation in training carries speed (RQ4). Removing rule regularisation, configuration randomisation or entropy annealing from the learned policy costs hundreds of steps or completion, and removing the recovery mechanism from the rule costs a third of its completions (RQ5). The learned policy's heavy right tail resists training-side remedies but is removed by applying the rule's recovery step at inference. The lookahead rule, at one to six minutes per instance, is 13–25% faster than both online controllers on every continuous configuration we measured, so the learned policy's case rests on latency and scaling, not on makespan.

Contributions. (i) A geometric chain-relay rule with deadlock recovery and a fleet-size-agnostic learned policy for connectivity-constrained delivery, both built on a projection that makes the constraint hold by construction. (ii) A paired evaluation across fleet sizes that shows the two controllers are complementary, together with the mechanism behind the boundary, and evidence that the learned policy transfers upward

to fleets three times larger than any it was trained on. (iii) A set of eleven documented negative results on closing the relay-limited deficit by changing the learner, and a sensitivity result showing that half of the deficit is attributable to the enforcement model. (iv) A space–time network model of the problem, solved exactly on small instances, with an evaluation that runs every method in the model's own decision space so that its optimum is a valid bound, and an explicit account of what each measurement does and does not say. (v) Two learning-free methods that calibrate the headroom: a rollout lookahead on the rule that is the strongest online-capable controller we have on makespan, and a beam search in the model's state space that returns the proven optimum on 46 of 50 instances in under three minutes.

2. Related Work

2.1. Vehicle routing with drones

The problem belongs to the family of vehicle routing problems [21, 22], with a homogeneous capacitated fleet, a single depot, unit demands and a makespan objective; the makespan (or min-max) objective, as opposed to total distance, is the standard choice when the goal is to finish service as early as possible [20, 23]. Drone routing began with the flying-sidekick travelling salesman problem, in which a truck and a drone cooperate [6], and was extended to multiple drones [24], to exact and approximate solution methods [25, 26, 17, 27], to vehicle routing with drones [28, 29, 30] and to drone-only fleets with energy and payload constraints [31]. The surveys of Otto et al. [7], Macrina et al. [8] and Chung et al. [9] map this literature. Across all of it communication is free: a drone is never required to be reachable from the depot, and no vehicle gives up its delivery role to serve as a relay. Our problem adds exactly that requirement, and the main consequence for modelling is that routes are no longer sufficient as decision variables, because the constraint concerns where all vehicles are at every instant (Section 3.1). Dynamic vehicle routing [32] shares the need to decide online as the state evolves, which is the mode in which both of our controllers operate.

2.2. Drones in disaster and humanitarian logistics

Drones have been proposed for last-mile distribution when ground infrastructure is damaged. Chowdhury et al. [1] design a drone-based relief distribution network with a continuous approximation model; Rabta et al. [2] optimise a drone fleet for last-mile delivery under payload and range limits; Kim et al. [4] and Ghelichi et al. [5] plan drone fleets for medical deliveries in rural and urban settings; and Erdelj et al. [3] survey the roles drones can take in disaster management, including that of temporary communication nodes. These works plan at the strategic or tactical level (fleet size, depot location, route assignment) and assume that the control link exists. Our contribution is at the operational level: given a fleet and a depot, how the drones should move so that the link is maintained by the fleet itself while it delivers.

2.3. Aerial relays and connectivity maintenance

Using UAVs as communication relays is a mature topic in wireless networking: surveys cover flying ad-hoc networks [33], UAV communication issues [34] and UAV-enabled wireless networks [11]; throughput-optimal relay trajectories are derived in [10]; and relay placement for maintaining a link to a surveillance UAV is solved exactly in [12]. Multi-hop chains of mobile relays are studied in the robotics literature as communication chains [14], robotic routers [35] and relay positioning for team missions [36]. Connectivity maintenance for a team of mobile robots has a control-theoretic literature based on the algebraic connectivity of the communication graph [37, 13], and learned connectivity-preserving controllers exist [38]. Learning-based relay control has been applied to packet routing in UAV networks [39], to joint trajectory and power control of relay and mission UAVs [40] and to aerial relay networking [41]. In all of these the relays are the mission, or a dedicated sub-fleet with its own reward; they do not carry cargo, do not reload and do not minimise a delivery makespan. Switching between a task role and a relay role is an established idea in multi-robot exploration [42, 43], where the relay robots land and act as fixed nodes; our drones switch roles repeatedly and the relay positions move with the lead drone.

2.4. Time-expanded network models and symmetry

Our exact model is a time-indexed formulation on a space–time network. Time-expanded networks originate with Ford and Fulkerson's flows over time [15]; Skutella [16] surveys the theory. The dependence of such models on the time discretisation is well known, and dynamic discretisation discovery [44] refines the time grid where needed; we discretise space as well, which is the usual device in MILP trajectory planning for aircraft [45]. Connectivity is encoded as a single-commodity flow per period whose feasibility is equivalent to the existence of a multi-hop chain (Section 3.1). The drones are identical, so the model carries the usual index symmetry, which we break with the ordering constraints of Sherali and Smith [46]; Margot [47] surveys symmetry handling in integer programming.

2.5. Learning for routing and cooperative control

Neural solvers for routing problems, from pointer networks [48] to attention-based constructive policies [49, 19], show that set-based encoders with attention generalise across instance sizes; Bengio et al. [50] and Mazyavkina et al. [51] survey machine learning for combinatorial optimisation. Learned solvers for the min-max multiple travelling salesman problem [20, 52] are the closest in objective to ours, and we borrow the size-agnostic set encoding; none of them has a connectivity constraint, and none decides online. Dual-mission fleets in which the same drones deliver and sense [53] formalise a per-delivery detour budget, but the secondary task there is sensing rather than communication and it does not couple the vehicles' positions. On the learning side we build on soft actor-critic [18], centralised critics for decentralised execution [54, 55], value decomposition [56, 57], cooperative multi-agent policy optimisation [58] (see [59] for a review), temporal abstraction through

options [60, 61, 62], invalid-action masking [63] and safe-action projection [64]. Our negative results in Section 6.3 test several of these ideas directly: per-agent credit, joint-action critics, monotonic value mixing and potential-style shaping [65] are all tried as remedies for the relay-limited deficit and none of them removes it.

2.6. Position of this work

To the best of our knowledge, no prior work combines (i) a homogeneous fleet carrying physical cargo to many demand points with capacity and reloading, (ii) a hard requirement that every drone be connected to the depot by a multi-hop chain at every step, and (iii) a makespan objective. In this setting relay duty competes with cargo throughput and the fleet must return to the depot repeatedly, which is what makes role assignment a sequential decision rather than a one-off placement. We contribute an exact space–time network model of this problem, a constructive online rule with a deadlock-recovery mechanism, and a learned size-agnostic policy, and we report where each is preferable.

3. Problem Statement

A depot sits at $c \in [0, 1000]^2$ metres and M demand points $d_1, \ldots, d_M$ lie within $0.9\,NR$ of it, where N is the number of drones and $R = 300$ m the communication range. Each drone flies at $v = 13.89$ m/s and carries $Q = 5$ units. Every delivery and every reload incurs a *service time*: unloading one unit at a demand point takes $S = 10$ steps and reloading to Q at the depot takes S steps, and during these steps the drone is stationary. A demand point or the depot is reached when a drone is within 25 m of it, and one step is one second. The service times matter for the connectivity constraint below: a stationary drone cannot move to keep a link alive, so its teammates must keep the chain intact around it. Every demand point needs exactly one unit. Drones do not collide, are not affected by wind, and battery is not modelled.

Connectivity. Let G_t be the graph whose vertices are the depot and the N drones and whose edges join vertices within R of each other. The fleet is *connected* at step t if G_t is connected, which is the same as saying that every drone has a multi-hop path to the depot. The constraint is that G_t is connected at every t. It is enforced rather than penalised (Section 4.1).

Objective. The makespan is the step at which the last demand point is served. A controller *completes* an instance if it serves all demand points within a budget of 10,000 steps. We report completion rate and, over completed instances, makespan.

Configurations. A configuration is a triple of fleet size N, number of demand points M and depot placement (fixed at the map centre, or uniformly random). The *standard* configuration is $N = 4$, $M = 50$ with the fixed depot. The learned policy is trained with $N \in \{3, 4\}$, $M \in [20, 80]$ and a random depot; all other configurations are unseen by it. The rule has no training.

3.1. A space–time network model, exact on the discretisation

The third element we propose, besides the two controllers, is an optimisation model of the problem. It serves two purposes: it states the problem without ambiguity, and on small instances it provides a plan obtained by exact optimisation against which both controllers are measured.

Why the model takes this form. Vehicle-routing formulations of drone delivery [6, 25] assign routes to vehicles and never need to know where a vehicle is at a given instant. The connectivity requirement of Section 3 is different in kind: it is a constraint on the *positions of all drones at every step*, because the communication graph G_t is built from those positions. A model must therefore carry every drone's position over time, and the natural object for that is a *time-expanded network* [15]: the set of positions is copied once per time period and arcs join a position in one period to the positions reachable in the next. Positions in the plane are uncountable, so space is discretised as well, to a lattice; the result is a *space–time network* on which drones are unit flows, and a time-indexed MILP on this network is exact for the discretised problem. Connectivity is then a property of which lattice nodes are occupied in each period, and we encode it with a single-commodity flow per period that is feasible exactly when the occupied nodes form a multi-hop chain to the depot. The model is *exact* in the sense that its optimum is the true optimum of the discretised problem; its relation to the continuous problem is stated at the end of this section.

Discretisation. A period is $\kappa = S = 10$ steps, so a service occupies exactly one period and a manager decision of the learned policy spans two; the per-period flight range is $\ell = v\kappa = 138.9$ m. Space is a square lattice G of spacing $h = \ell/\sqrt{2} = 98.2$ m over the bounding box of the instance, from which nodes farther than $0.9NR$ from the depot are deleted because no chain can reach them; the depot and every demand point are added as exact nodes, so the interaction test is exact there. Figure 1 shows the resulting node set for a reference instance. With this spacing all eight king moves are feasible in one period; the move set and the communication set are

$$A(i) = \{j \in N_{\mathrm{v}} : \|p_i - p_j\| \le \ell\} \ni i, \tag{1}$$

$$E = \{(i, j) \in N_{\mathrm{v}}^2 : i \ne j,\ \|p_i - p_j\| \le R\}. \tag{2}$$

Axis-aligned moves cover h rather than ℓ per period, so the lattice slows a drone by up to 29% in those directions; this discretisation error always acts against the model. The lattice must also be fine enough to hold a relay chain: serving a demand point at distance d needs the intermediate drones placed within the chain's slack $NR - d$, which requires roughly $h \lesssim (NR - d_{\max})/(N - 1)$; a 196 m lattice makes the reference instances infeasible although they are feasible in the plane, and 98 m does not.

Node pruning by the convex hull. Most lattice nodes inside the reach limit are never useful. Let H be the convex hull of the depot and the demand points. For any configuration of the fleet, projecting every drone that lies outside H onto H does not increase any pairwise distance, because projection onto a convex

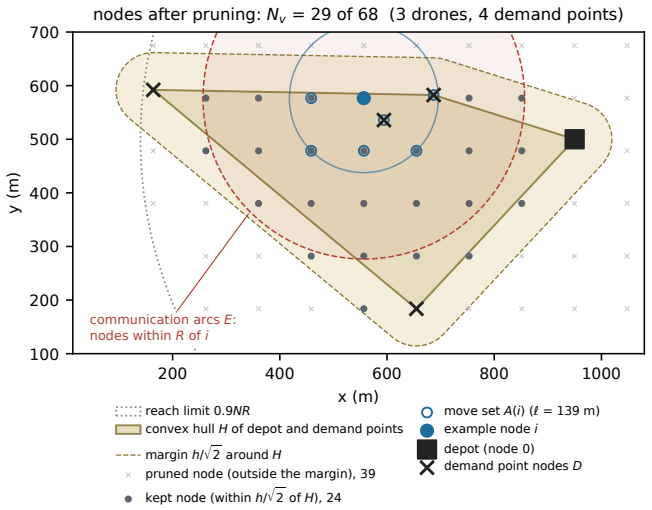


Figure 1: Nodes of the space–time network for a reference instance with three drones and four demand points: the depot (node 0), the demand-point nodes D, and the lattice G of spacing $h = 98.2$ m clipped to the reach limit $0.9NR$ (dotted). The shaded polygon is the convex hull H of the depot and the demand points and the lighter band is the margin of width $h/\sqrt{2}$ around it; lattice nodes inside the band or the hull are kept (dots) and those outside are pruned (crosses), leaving 29 of 68 nodes. For one example node i the figure marks its move set $A(i)$ (the nodes reachable in one period, $\ell = 138.9$ m) and its communication disc of radius R, which defines its arcs in E.

set is non-expansive; hence it preserves every communication link, every one-period move and every service, and the projected plan has the same makespan. The continuous problem therefore always has an optimal plan inside H, and relay positions outside the hull can be discarded without loss. On the lattice we keep the nodes within $h/\sqrt{2}$ of H, the half-diagonal of a cell, so that every point of H still has a kept node within one lattice step; this margin guards against the case in which the projection of a lattice node is not itself a lattice node. The pruning removes 45% of the nodes of the 34 reference instances on average (from 2,881 to 1,594 in total) and, on the instances where we can check, leaves the lattice optimum unchanged (Section 6.2). Figure 1 shows the pruned and kept nodes.

Notation. Table 1 lists the sets, parameters and decision variables. Drones are identical, so a drone index only distinguishes units; the symmetry-breaking cut at the end of the model exploits this. The model is given in (3)–(16).

Reading the model. The objective (3) minimises the makespan C; the second term is a tie-break with weight $\varepsilon\bar{T}N \ll 1$ that prefers plans keeping idle drones at the depot and cannot change the optimal makespan.

Position and movement, (4)–(6). Every drone is at exactly one node at every period boundary and starts at the depot. A drone may be at node i at boundary $t+1$ only if at boundary t it was at i itself or at a node from which i is reachable within one period. Because every move takes exactly one period, the position of every drone is defined at every boundary, which is what the connectivity constraints below need.

Deliveries, cargo and reloading, (7)–(12). Each demand point is served exactly once (7). A delivery during period t requires the drone to be at the demand point at both ends of the period (8), which is the service time of Section 3: the drone stands still for one period. Reloading is likewise a full period at the depot (9). A drone does at most one service per period and can only deliver what it carries (10); cargo is tracked by (11) starting full; and a reload fills the drone completely (12), because ρ_{kt} can be positive only in a reload period and the next cargo must then be Q.

Multi-hop connectivity, (13)–(14). These two families are the part that distinguishes this model from drone-routing models. In every period, every occupied node ships one unit of flow, the depot is the only sink, and an arc may carry flow only if it leaves an occupied node. The system is feasible if and only if every occupied node is joined to the depot by a path whose intermediate nodes are occupied and whose consecutive distances are at most R, that is, if and only if the simulator's multi-hop connectivity test succeeds at that period boundary. The proof is short: if such paths exist, routing each node's unit along its path satisfies both families; conversely an unoccupied node $i \neq 0$ has $g_{ijt} = 0$ on all its outgoing arcs by (14) and therefore no incoming flow by (13), so all flow travels inside the occupied set and, since the depot is the only node without a conservation row, a flow decomposition yields the required paths. The mirror-image capacity rows $g_{ijt} \leq Nu_{jt}$ are implied and omitted, which halves the dominant row count. Note what the constraint does to the problem: a drone that has nothing to deliver cannot simply wait at the centre if a teammate needs it as a hop, and a drone cannot fly to a demand point that would leave it farther than R from every occupied node. Relaying is therefore not a separate task but a consequence of where the fleet is.

Makespan and symmetry, (15)–(16). The makespan is at least the period of every delivery. Because drones are identical, any plan can be relabelled so that drone k serves no demand point with index below k; (16) imposes this ordering and removes no objective value.

What the optimum means. The optimum of the model is exact for the discretised problem. For the continuous problem of Section 3 it is an *upper bound*: drones are confined to lattice nodes and axis moves are shortened, so every lattice plan is a feasible continuous plan but not conversely, and a continuous controller can therefore be faster than the lattice optimum. The solver's dual bound is a bound on the lattice problem only. A certified lower bound through a cell relaxation, in which a lattice node stands for its cell and communication is tested by the minimum distance between cells, was attempted and abandoned: it inflates a three-hop chain's reach from 900 m to 1,317 m and returns bounds below the trivial analytic bound. We therefore

Table 1: Notation of the space–time network model.

Symbol	Domain	Meaning
Sets		
$N_{\mathrm{v}} = \{0\} \cup D \cup G$		nodes: the depot (index 0), the demand-point nodes D, the lattice nodes G
K	$\lvert K\rvert = N$	drones
$T = \{0, 1, \ldots, \bar{T}\}$		period boundaries; period t runs from boundary t to $t+1$
$A(i) \subseteq N_{\mathrm{v}}$		nodes reachable from i within one period, including i (move set)
$E \subseteq N_{\mathrm{v}}^2$		ordered pairs of distinct nodes within communication range R (communication arcs)
Parameters		
Q	5	cargo capacity (units)
R	300 m	communication range
κ, ℓ, h	10 steps, 138.9 m, 98.2 m	period length, flight range per period, lattice spacing
$\bar{T}$		horizon in periods, set to 1.15× the rule's makespan
ε	10^{-5}	tie-breaking weight
$\mathrm{idx}(j)$		index of demand point j in a fixed ordering of D
Decision variables		
y_{ikt}	$\{0, 1\}$	1 if drone k is at node i at boundary t
s_{jkt}	$\{0, 1\}$	1 if drone k delivers to demand point j during period t
r_{kt}	$\{0, 1\}$	1 if drone k reloads at the depot during period t
q_{kt}	$\{0, \ldots, Q\}$	cargo on board drone k at boundary t
ρ_{kt}	$[0, Q]$	cargo taken on by drone k during period t
g_{ijt}	$[0, N]$	connectivity flow on arc (i, j) in period t
u_{it}	$\{0, \ldots, N\}$	occupancy of node i at boundary t, $u_{it} = \sum_k y_{ikt}$, with $u_{0t} := 1$
C	$[0, \bar{T}]$	makespan in periods

$$\min \; C + \varepsilon \sum_{i \in N_{\mathrm{v}} \setminus \{0\}} \sum_{k \in K} \sum_{t \in T} y_{ikt} \tag{3}$$

$$\text{s.t.} \quad \sum_{i \in N_{\mathrm{v}}} y_{ikt} = 1 \qquad \forall k \in K,\ t \in T \tag{4}$$

$$y_{0k0} = 1 \qquad \forall k \in K \tag{5}$$

$$y_{i,k,t+1} \le \sum_{j \in A(i)} y_{jkt} \qquad \forall i \in N_{\mathrm{v}},\ k \in K,\ t < \bar{T} \tag{6}$$

$$\sum_{k \in K} \sum_{t < \bar{T}} s_{jkt} = 1 \qquad \forall j \in D \tag{7}$$

$$s_{jkt} \le y_{jkt}, \qquad s_{jkt} \le y_{j,k,t+1} \qquad \forall j \in D,\ k \in K,\ t < \bar{T} \tag{8}$$

$$r_{kt} \le y_{0kt}, \qquad r_{kt} \le y_{0,k,t+1} \qquad \forall k \in K,\ t < \bar{T} \tag{9}$$

$$\sum_{j \in D} s_{jkt} + r_{kt} \le 1, \qquad \sum_{j \in D} s_{jkt} \le q_{kt} \qquad \forall k \in K,\ t < \bar{T} \tag{10}$$

$$q_{k,t+1} = q_{kt} - \sum_{j \in D} s_{jkt} + \rho_{kt}, \qquad q_{k0} = Q \qquad \forall k \in K,\ t < \bar{T} \tag{11}$$

$$\rho_{kt} \le Q\, r_{kt}, \qquad q_{k,t+1} \ge Q\, r_{kt} \qquad \forall k \in K,\ t < \bar{T} \tag{12}$$

$$\sum_{j:(i,j) \in E} g_{ijt} \; - \sum_{j:(j,i) \in E} g_{jit} = u_{it} \qquad \forall i \in N_{\mathrm{v}} \setminus \{0\},\ t \in T \tag{13}$$

$$g_{ijt} \le N\, u_{it} \qquad \forall (i, j) \in E,\ t \in T \tag{14}$$

$$C \ge t\, s_{jkt} \qquad \forall j \in D,\ k \in K,\ t < \bar{T} \tag{15}$$

$$s_{jkt} = 0 \qquad \forall k \in K,\ j \in D \text{ with } \mathrm{idx}(j) < k,\ t < \bar{T} \tag{16}$$

call the solver's value a *reference plan*, replay every plan in the continuous simulator, and report the replayed makespan.

4. Proposed Controllers

Figure 2 summarises the setting and the two controllers. Both controllers decide at the level of *goals*: at every decision point each drone is given a goal (a demand point, the depot, or a

relay slot) and flies towards it in a straight line at full speed until the next decision point. The two controllers differ only in how goals are chosen; the option set, the low-level flight, the relay-slot geometry and the connectivity projection are shared and are described first.

4.1. Shared interface: options, relay slots, projection

Decision points and options. A decision point occurs every 20 steps, and additionally as soon as any drone's goal becomes invalid: its demand point has been served by another drone, it has run out of cargo while heading to a demand point, it has become full while heading to the depot, or its relay slot has disappeared because the chain it belonged to shortened. At a decision point every drone receives one of $K = 7$ options: one of its five *candidate demand points*, which are the five unserved demand points nearest to it; *return* to the depot; or *relay*. An option is *valid* for a drone only if it makes sense at that moment: a demand point option requires cargo on board and an existing candidate, return requires the drone not to be full, and relay requires a free relay slot (defined next) and at least one teammate that is not itself relaying. Invalid options are masked for the learned policy and never produced by the rule; a drone with no valid option returns to the depot.

Relay slots. The environment, not the controller, decides where a relaying drone stands. For drone i the *frontier* f_i is the point farthest from the depot among the delivery goals of the other drones and the current positions of the other non-relaying drones (if there is none, the farthest unserved demand point). A chain to f_i at distance d needs $n = \min(\lceil d/(0.9R) \rceil - 1,\ N - 1)$ relay slots, placed on the segment from the depot to f_i at the fractions $1/(n+1), \ldots, n/(n+1)$ of its length; the factor 0.9 leaves slack so that a drone standing slightly off its slot does not break the link. Drone i takes the free slot nearest to it. A drone whose slot disappears holds position until the next decision point. The controllers therefore decide only *whether* a drone relays; the slot follows from the fleet's geometry.

Low-level flight and connectivity projection. Every step, each drone commands a displacement of length v towards its goal (zero while it is serving). Enforcing connectivity by a reward penalty would punish situations that no action can prevent, for example a drone that is stationary while unloading and is stranded when a teammate departs; connectivity is therefore enforced by projection. If the commanded positions keep G_t connected they are applied. Otherwise the projection repeatedly selects one moving drone, scales its displacement down to the largest factor in $[0, 1]$ under which the fleet is connected (found by bisection with the other drones' current scalings fixed), and stops as soon as the fleet is connected; at each round it selects the drone whose restriction leaves the most drones connected and breaks ties at random, so that no single anchor drone is starved. The previous positions are always connected, so zero scaling is a valid fallback and the constraint holds by construction. A drone whose displacement is scaled below 10% is *stalled* for that step, and a *partial deadlock* is a step in which all but at most one of the moving drones are stalled; the number of consecutive partial-deadlock steps, δ_t, is what the rule's recovery mechanism monitors. This projection scales a displacement along its own direction; Section 6.3 studies a variant that keeps the tangential component.

4.2. Geometric chain-relay rule with deadlock recovery

The rule is a deterministic rule evaluated at every decision point (Procedure 1). It first gives every drone a default: a drone with cargo goes to its nearest unserved demand point and a drone without cargo returns to the depot. It then selects the *lead*, the drone with cargo that is farthest from the depot, computes how many hops the lead's target needs, and turns that many of the remaining drones, the farthest ones first, into relays. The number of relays is adaptive and the relays are the drones already closest to where the chain must stand, which is what makes the rule cheap: it costs $O(N \log N)$ per decision and 0.05–0.12 ms in our implementation.

Deadlock recovery. Because the rule is deterministic, a geometry that stalls the fleet makes it re-issue the same assignment at every decision point and the stall persists; without an recovery mechanism the rule completes 62% of standard instances. The recovery mechanism is a timed override. It *starts* at the first decision point at which the fleet has been in partial deadlock for at least k_{stall} consecutive steps and no recovery is in progress. It *lasts* for h_{hold} decision points, that is, roughly $20\,h_{\text{hold}}$ steps, fewer when goals become invalid early; during these decision points the rule is evaluated as usual and then overridden as in step 6. It *ends* when the count e reaches zero, after which the plain rule resumes; if the deadlock persists it can start again at the next decision point. Two modes are implemented. In *lead swap* the current lead is sent back to the depot and the role of lead passes to another drone; on the m-th recovery of an episode it passes to the m-th farthest drone, so that repeated recoveries cycle through the fleet rather than swapping the same two drones. In *retreat* every drone except the lead returns towards the depot, which slackens the chain. Section 6.5 selects the mode and the two parameters on tuning seeds; lead swap with $k_{\text{stall}} = 120$ and $h_{\text{hold}} = 20$ is used in every comparison.

4.3. Rollout lookahead on the rule

The rule's weakest decision is the choice of target, which is made independently per drone and myopically. The *lookahead rule* keeps Procedure 1 as a base policy and replaces that choice by a one-step rollout [66]: at a decision point it enumerates a set of candidate assignments, simulates each to completion with the base rule, and commits to the assignment whose simulated makespan is shortest, then re-evaluates at the next decision point. The candidate set is the rule's own assignment; for each drone with cargo and each of its nearest candidates, the same assignment with that drone sent to that demand point; for each drone with cargo but not full, the same assignment with that drone sent to reload now; the product of per-drone choices among the two nearest demand points, relaying and reloading, restricted to distinct targets; and, for every candidate, two relay modes, the slot of Procedure 1 or a *follow* mode in which

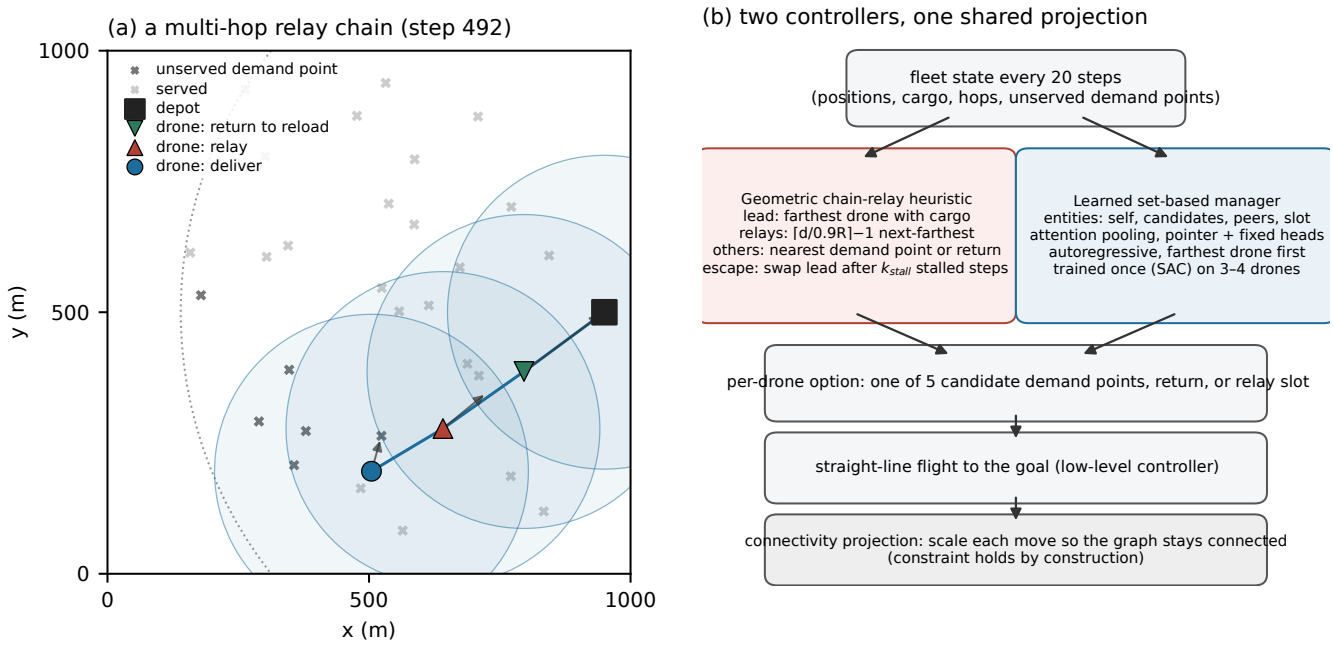


Figure 2: (a) An instance with three drones under the rule at step 499: the lead (blue circle) heads for a far demand point, one drone relays on the segment to the depot (red triangle), and one drone returns to the depot to reload (green triangle); shaded discs are communication ranges and the dotted circle is the reachable region $0.9NR$. (b) The two controllers emit the same per-drone options and share the straight-line low-level controller and the connectivity projection.

Procedure 1. Geometric chain-relay rule with deadlock recovery, run at every decision point. State kept between calls: remaining recovery length e (initially 0) and recovery count m (initially 0); both reset at the start of an episode. Parameters: $k_{\text{stall}} = 120$, $h_{\text{hold}} = 20$, mode $\in$ {lead swap, retreat}.

1. **Recovery trigger.** If the fleet has been in partial deadlock for $\delta_t \geq k_{\text{stall}}$ consecutive steps and $e = 0$, set $e \leftarrow h_{\text{hold}}$ and $m \leftarrow m + 1$.
2. **Default assignment.** For each drone i: if it has no cargo, option $\leftarrow$ return; else if it has no unserved candidate, option $\leftarrow$ relay; else option $\leftarrow$ its nearest unserved demand point.
3. **Lead.** If no demand point is unserved, stop. Otherwise let $r_i = \|p_i - c\|$ and let the lead be the drone with cargo that maximises r_i (if no drone has cargo, the drone maximising r_i).
4. **Relay count.** Let $d = \max(r_{\text{lead}}, \|d_{\text{goal}} - c\|)$ where d_{goal} is the lead's nearest unserved demand point, and $n = \min(\max(1, \lceil d/(0.9R) \rceil) - 1,\ N - 1)$.
5. **Relay identity.** Sort the drones other than the lead by r_i in decreasing order and set the option of the first n of them to relay.
6. **Recovery manoeuvre** (only if $e > 0$; then $e \leftarrow e - 1$). *Lead swap*: sort all drones by r_i in decreasing order, let the new lead be the $\min(m, N-1)$-th drone of that order (0-based), set the old lead's option to return and, if the new lead has cargo and an unserved candidate, set its option to its nearest unserved demand point. *Retreat*: set the option of every drone except the lead to return.
7. Return the options; the environment converts them into goals and assigns relay slots.

a relay's goal is the lead's goal and the connectivity-preserving mover holds it at the farthest reachable point of the chain, so that relays advance with the lead instead of parking on a slot. The follow mode works only with the lattice mover of Section 6.2, which chooses the best connected joint move; in the continuous space the per-step projection lets a following relay and its lead block each other, so there the candidate set omits it and keeps the slot geometry. Rollouts are capped at 1,500 steps and an unfinished rollout is scored by its length plus 50 steps per unserved demand point. The lookahead needs the simulator and the instance, and costs tens of seconds to minutes per instance rather than milliseconds per decision; it is a rule in the sense that it involves no training and no search over a solution representation, and it is included because a baseline that is cheap to build should be as strong as cheap methods allow.

4.4. Search heuristics over visit orders and over task sequences

The rule decides myopically: at every decision point the lead goes to its nearest unserved demand point. The third method keeps the rule's execution layer and replaces that choice by an offline search over a solution representation, evaluated by simulation, so the objective needs no model of the dynamics and automatically respects the connectivity constraint. We use two representations of increasing expressiveness.

Visit-order search. A solution is a *priority list* of demand points per drone. At a decision point a drone's goal is the first demand point on its own list that is still unserved, or the depot if its list is exhausted or it is empty; relays and the deadlock recovery are assigned exactly as in Procedure 1, steps 3–6. The initial solution is read off a run of the rule, by recording which drone served which demand point in which order, so the search starts from the rule's behaviour. Neighbours relocate one demand point to a random position in a random list, swap two demand points between lists, or reverse a segment of one list. We compared simulated annealing, variable neighbourhood search, iterated local search and GRASP under an equal budget of 1,200 simulations on the instances of Section 6.2; their mean gaps to the optimum lie within 0.8 percentage points of each other and no pair differs significantly (Wilcoxon $p \geq 0.06$), so the choice of metaheuristic is immaterial and we use iterated local search.

Task-sequence search. The visit-order search still leaves relaying to the rule's geometry: a relay stands on a slot of the segment to the current frontier and waits. The second representation makes relaying itself a search variable. A solution is a sequence of *tasks* per drone, each either *deliver j*, *relay at node p until demand point j is served* (by whichever drone serves it), or *reload*. A drone executes its sequence in order, skipping tasks whose demand point has already been served and inserting a reload when it is empty, and flies to each task's node with the same connectivity-preserving mover as the other methods. The initial sequence is again read off a run of the rule, now recording where and for how long each drone relayed. Neighbours add the visit-order moves and four relay moves: insert a relay task for a delivery at a node near the chain's slot for that delivery, delete a relay task, move a relay to an adjacent lattice node, and change the demand point a relay waits for. A fleet that makes no progress for six periods rotates the stalled drones' current tasks to the end of their sequences.

Beam search in the model's state space. The third search abandons the controllers' option interface altogether and searches the state space of the space–time model directly, with the same transition set as the exact solver of Section 6.2: a state is the multiset of (node, cargo) pairs of the drones and the set of served demand points, a period moves every drone to a node of its move set subject to connectivity at both ends of the period, and staying on a demand point with cargo delivers, staying at the depot reloads. The exact solver expands every reachable state per period; the beam search keeps the B states with the smallest estimated remaining time, where the estimate is the largest over unserved demand points of the minimum number of periods some drone needs to reach it with cargo, plus the sum of those numbers and of the deliveries still due divided by N, plus, when the unserved demand points outnumber the cargo on board, the depot round trips that the shortfall implies. Ties are broken at random. The result is a feasible plan of the model, so it can never beat the optimum, and the search approaches the exact solver as B grows; it is replayed in the continuous simulator to confirm connectivity. All three searches are offline: they need the instance in advance and spend seconds to an hour on it, so we report them as quality references rather than as controllers.

4.5. Hierarchical learned policy

The learned controller is hierarchical (Figure 3). The high level, a *manager*, acts only at decision points and outputs one option per drone; the low level is the shared straight-line flight of Section 4.1, so the manager's choices are held for 20 steps and the problem it solves is a semi-Markov decision process over options [60]. Only the manager is learned. Separating the levels has two consequences that the rest of the design relies on: the manager sees the fleet at a coarse time scale at which a formation can be held, and the low level, being a fixed rule, can be reasoned about and projected for connectivity without learning.

Set-based encoder. The manager encodes drone i from four kinds of entities, all distances divided by R. *Own state*, 10 features: cargo fraction; distance to the depot; hop depth divided by 8; fraction of demand points unserved; elapsed time divided by the budget; its realised-movement ratio (an exponential average of achieved over commanded displacement); whether it could advance one full step towards its goal alone without disconnecting the fleet; the fleet-wide realised-movement ratio; service timer; and whether it is a cut vertex of G_t. *Candidate demand points*, five entities of 6 features: distance; the two components of the unit direction vector; whether a teammate already targets it; its distance from the depot; and the hops its chain requires, divided by 8. *Peers*, $N-1$ entities of 9 features: distance; the two components of the relative position vector; cargo fraction; hop depth divided by 8; and a four-way one-hot of the peer's current goal kind (deliver, return, hold, relay). *Relay slot*, 4 features: distance; the two components of the unit direction vector; and the number of slots in the current chain, divided by 8. Each entity type passes through a two-layer MLP of width 128. The drone's own embedding is the query of two multi-head attention layers (4 heads) that pool the peer set and the candidate set into one vector each; the four vectors are concatenated and fused by a further MLP into the drone's embedding h_i. A *pointer head* scores each candidate from $[h_i; e_{\text{cand}}]$ and a *fixed head* scores return and relay from h_i; the seven scores are masked and normalised by a softmax. Nothing in the network depends on N or M, which is why one set of weights runs on fleets it never saw.

Autoregressive decision. The N options are not sampled independently. Drones are ordered by distance from the depot, farthest first; the manager samples the first drone's option, writes its new role into the role one-hot of the peer features that the remaining drones see, re-encodes, samples the next drone, and so on (Procedure 2). A formation of one lead and k relays arises only by coincidence under independent sampling but can be composed deliberately under sequential decisions; the cost is N sequential network calls per decision point.

Training. The manager is trained with discrete soft actor-critic [18] on option transitions. A transition stores the fleet observation at a decision point, the N options, the option return $\sum_\tau \gamma^\tau r_\tau$ accumulated over the steps until the next decision point, the observation at that point, and the discount $\gamma^{\Delta t}$ with $\gamma = 0.997$ per step. The team reward is $+10$ per delivery, $+30$ per reload, $-0.01N$ per step and a residual penalty of $-0.1N$ times the fraction of demand points still unserved. A twin-Q critic with attention across the drone embeddings outputs per-drone action values whose mean over drones is regressed on the option return; the actor minimises the standard expectation form $\mathbb{E}_{a\sim\pi}[\alpha \log \pi(a) - Q(a)]$ per drone, with teacher forcing of the autoregressive order. Three elements of the training configuration matter and are ablated in Section 6.5. (i) The entropy target is annealed linearly from $0.6 \log n_{\text{valid}}$ to $0.25 \log n_{\text{valid}}$ over the first 400 episodes: a high target keeps the policy random enough to recovery deadlocks early in training, a low one lets it hold a multi-step formation later. (ii) A cross-entropy term

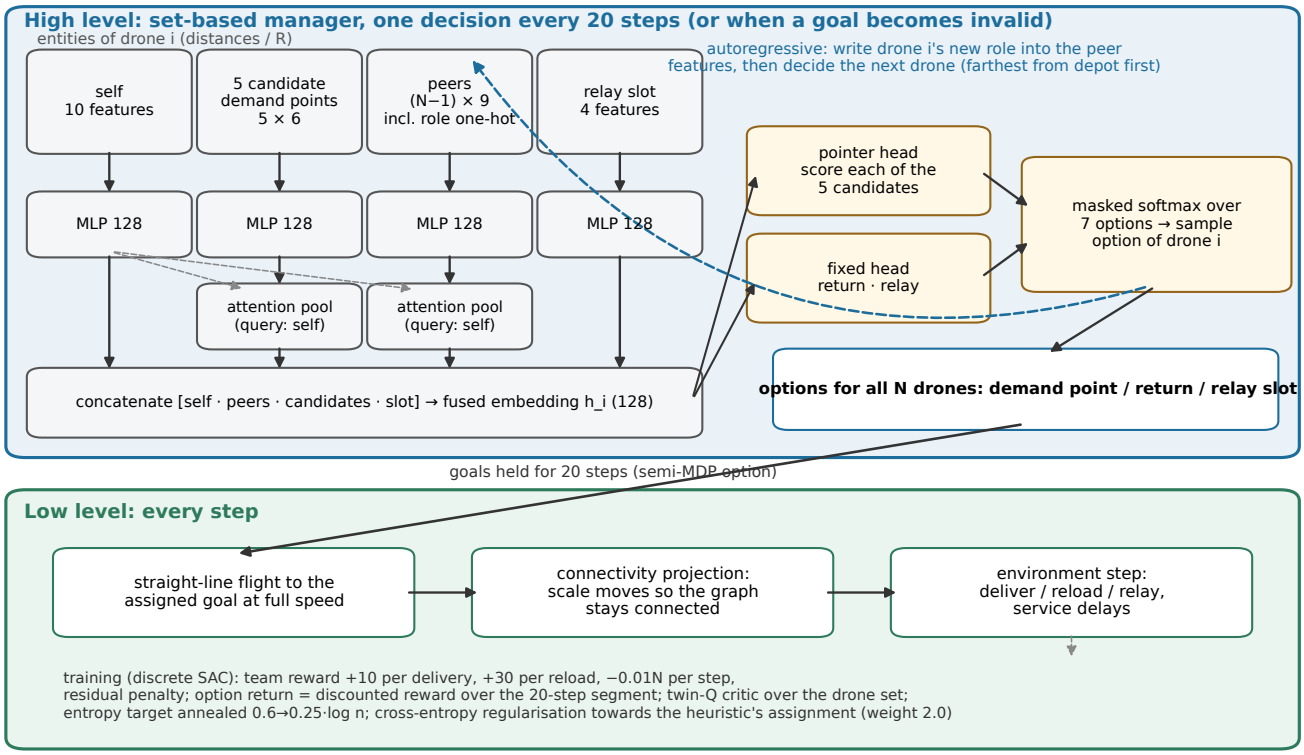


Figure 3: The hierarchical learned controller. High level: the set-based manager encodes each drone from four kinds of entities, pools the peer and candidate sets with attention queried by the drone's own embedding, and scores the five candidates with a pointer head and return/relay with a fixed head; drones are decided one at a time, farthest from the depot first, and each decision is written into the peer features seen by the next drone. Low level, every step: straight-line flight to the assigned goal, the connectivity projection, and the environment step. Options are held for 20 steps.

Procedure 2. Learned manager at a decision point (inference).

1. Build the entity observation of every drone from the fleet state; compute the validity mask of the seven options for every drone (Section 4.1).
2. Order the drones by distance from the depot, farthest first.
3. For each drone i in that order: encode all entities with the current peer roles, compute the seven masked logits of drone i, sample its option from the softmax (sampling, not argmax, is the deployed inference), and write the chosen role (deliver, return or relay) into the peer role one-hot of drone i as seen by the drones not yet decided.
4. Return the N options; the environment converts them into goals and assigns relay slots.

with weight 2.0 pulls the policy towards the option the rule's assignment rule (Procedure 1, steps 2–5, without recovery) would take, wherever that option is valid. (iii) Every training episode samples $N \in \{3, 4\}$, $M \in [20, 80]$ and a random depot. Optimisation uses Adam at 3×10^{-4}, batches of 256 transitions from a replay buffer of 60,000, one gradient step every 4 environment steps after a warm-up of 400 transitions, and Polyak averaging of the target critic with $\tau = 0.005$. Checkpoints are scored every 50 episodes on 80 validation instances with a 3,000-step budget by $100f + 40(1 - \bar{t}/3000)$, where f is the completion rate and $\bar{t}$ the mean steps; training stops after eight evaluations without improvement or at 1,000 episodes. One run takes about 13 hours on a single RTX 5080 when four runs share the GPU. The rollout lookahead of Section 4.3 can also be run with the learned policy as its base policy, so that the policy both proposes the candidate assignments and evaluates them by simulation; Section 6.2 reports this learning-plus-search combination.

5. Experimental Setup

Instances and configurations. Instances are generated from integer seeds; a seed, N, M and the depot mode determine the instance exactly, so two controllers can be run on identical instances. Evaluation seeds are disjoint from the seeds used for training, for checkpoint selection and for tuning the rule. We evaluate on the standard configuration ($N = 4$, $M = 50$, fixed depot) and on eight unseen configurations with random depots: $N/M \in \{3/20, 3/30, 4/80, 5/50, 6/80, 8/150, 10/300, 12/400\}$.

Baselines. Four learning-free baselines establish that the problem is not solved by naive assignment. All use the shared interface of Section 4.1 and are evaluated at the same decision points. *Greedy, no relay* is step 2 of Procedure 1 alone: every drone with cargo flies to its nearest unserved demand point and every empty drone returns; nobody ever relays, so the chain is maintained only by the projection, which stalls the fleet. *One (two) fixed relays* assigns the first one (two) drones of the fleet to relay permanently and runs the greedy rule on the others; the relay count is fixed in advance instead of adapting to the lead's distance. *Geometric rule, no recovery* is Procedure 1 without

step 6. The lookahead rule of Section 4.3 is run on the three main configurations in the continuous space with the lookahead evaluated every 100 steps or when a goal becomes invalid and three nearest candidates per drone, and on the exact instances in the model's decision space with the full candidate set. All learning-free controllers are deterministic except for the random tie-breaking of the projection.

Reference model. Instances with $M \leq 5$ demand points are solved to proven optimality with the breadth-first search of Section 6.2 on the hull-pruned lattice; instances with $M \geq 6$ are solved with the space–time MILP in Gurobi 13 [67], four threads and one hour per instance. Every optimal or best-found plan is replayed in the continuous simulator. The cargo-carrying exact solver used for the reload instances allows a drone to stand on a demand point with cargo without delivering, as the model does; an earlier version forced the delivery, which restricts the model, and re-solving all fifteen reload instances without that restriction changed none of the optima.

Paired protocol. Averaging only the episodes each method completes favours the method that gives up on hard instances, and a single long stall moves a 40-run mean by over 100 steps. All comparisons between the two controllers are therefore *paired*: both run on the same instance, the difference is taken over instances both complete, and we report the median paired difference, the win–loss count and a Wilcoxon signed-rank p-value. Inference samples from the policy; deterministic argmax inference has the same median but a far heavier tail. The standard configuration uses 480 runs (60 seeds, 8 repetitions, two independent measurement runs), the 3/30 and 5/50 configurations 200 (40 seeds, 5 repetitions) and the remaining configurations 80 (40 seeds, 2 repetitions). Repetitions differ because the projection's tie breaking is random. We found 40 runs too few to support any claim about tails.

6. Results

6.1. RQ1: Completion under the hard constraint

Connectivity was never violated in any training episode or evaluation run, as the projection guarantees. Table 3 reports completion within the 10,000-step budget. The learned policy completes every instance in every configuration. The rule with recovery completes 98–100%; its misses concentrate where drones are plentiful, where a long chain that is not needed still gets stationed and stalls. Without the recovery mechanism the rule completes 61.7% of standard instances, and the naive baselines complete almost none: the greedy assignment pulls the fleet apart, and fixed relay counts are wrong for most instances because the number of relays a demand point needs varies with its distance.

The lookahead rule. Table 2 adds the rollout lookahead of Section 4.3 on the same instances. It completes every instance, including the three on which the rule stalls, and it is faster than both online controllers on every configuration: 19% faster than the rule and 21% faster than the learned policy's median at 3/30, 25% and 22% at 4/50, 30% and 13% at 5/50, winning 38 to 59 of 40 to 60 paired instances against each. It pays for this with one to six minutes of simulation per instance, four to five orders of magnitude above the learned policy's decision time, and with the need for a simulator of the instance at hand. Fleet size no longer divides the space when this method is on the table: the question the rest of the paper answers is what the two millisecond controllers achieve, and Section 6.2 measures how far all of them are from the optimum.

Table 2: Mean makespan of the rule, the lookahead rule and the learned policy (per-instance median of its paired runs) on the three main configurations in the continuous space, with the number of completed instances or runs, wins–losses and Wilcoxon p of the lookahead rule against each, on instances all three complete, and its mean simulation time per instance.

config.	inst.	rule	lookahead	learned	vs rule		vs learned		time
					W–L	p	W–L	p	s/inst.
3/30	40	600 (40/40)	487 (40/40)	620 (200/200)	40–0	$< 10^{-4}$	39–1	$< 10^{-4}$	59
4/50	60	926 (59/60)	690 (60/60)	880 (480/480)	59–0	$< 10^{-4}$	59–0	$< 10^{-4}$	302
5/50	40	669 (39/40)	466 (40/40)	536 (200/200)	39–0	$< 10^{-4}$	38–1	$< 10^{-4}$	361

Table 3: Completion within the 10,000-step budget. Learning-free baselines on 120 (standard) or 40 (unseen) runs; the two proposed controllers on the paired run counts of Section 5. p is Fisher's exact test (two-sided) of each method's completion count against the learned policy on the same configuration.

Method	4/50 std.	p	3/30	p	5/50	p
Greedy, no relay	0%	$< 10^{-4}$	0%	$< 10^{-4}$	25%	$< 10^{-4}$
One fixed relay	0%	$< 10^{-4}$	2.5%	$< 10^{-4}$	40%	$< 10^{-4}$
Two fixed relays	3.3%	$< 10^{-4}$	52.5%	$< 10^{-4}$	65%	$< 10^{-4}$
Rule, no recovery	61.7%	$< 10^{-4}$	95%	0.027	92.5%	0.004
Rule + recov. (prop.)	473/480	0.015	200/200	1	191/200	0.004
Learned (prop.)	**480/480**	—	**200/200**	—	**200/200**	—
	3/20		4/80		6/80	
Rule + recov. (prop.)	80/80	1	80/80	1	79/80	1
Learned (prop.)	**80/80**		**80/80**		**80/80**	
	8/150		10/300		12/400	
Rule + recov. (prop.)	80/80	1	80/80	1	80/80	1
Learned (prop.)	**80/80**		**80/80**		**80/80**	

6.2. RQ2: Distance to the exact optimum of the space–time model

Exact solutions. We fix the lattice at $h = 98$ m, the spacing derived in Section 3.1, and solve the space–time model on it to proven optimality by breadth-first search over its state space. A state is the multiset of the three drones' nodes together with the set of served demand points; the fleet is identical so states are kept up to permutation, and the search expands all node triples reachable in one period that satisfy the multi-hop connectivity test, period by period, until every demand point is served. Because every transition is exactly a feasible column of the MILP, the first period at which the full set appears is the optimal makespan of the model. The search does not carry cargo, which relaxes the capacity constraint; we recover the optimal plan and check it, and on all forty instances no drone delivers more than $Q = 5$ units, so the relaxation is tight and the values are the optima of the full model. Two checks confirm the implementation: on the instance Gurobi could certify for $M = 4$ the search returns the same optimum in one second against 1,242 (seed 902, ten periods), and on the three instances with $M = 6$

that the MILP certified within an hour the two agree exactly (130, 100 and 120 steps). Each instance takes 0–11 seconds, so we solve ten instances for each of $M = 3, 4, 5, 6$. Connectivity is enforced on the lattice at both ends of every period: because the distance between two drones moving in straight lines is a convex function of time, an edge that holds at both boundaries holds throughout the period, so every plan of the model is connected at every instant of the continuous problem; a model that checks connectivity only at boundaries, as our first version did, lets 25 of 40 "optimal" plans break the chain mid-period by up to 100 m. To exercise reloading at a size the exact search can reach we also solve instances with capacity $Q = 2$, so that the fleet's 6 units fall short of $M = 7$ or 8 demand points and at least one reload is forced; here the search carries cargo in the state and takes 4–22 minutes per instance (ten instances with $M = 7$, five with $M = 8$).

Running the controllers in the model's decision space. A controller that flies freely and a plan confined to the lattice are not playing the same game, so a difference between them is partly an artefact of the discretisation. We therefore run the controllers in the model's own decision space. A drone occupies a lattice node at every period boundary and may move only to a node in $A(i)$; connectivity is checked at period boundaries, as in the model; a service occupies a whole period, so a drone that stands on an unserved demand point with cargo at a boundary delivers and stays there for that period, which is exactly $s_{jkt} = 1$; relay slots are snapped to the nearest node. The low-level mover is itself close to optimal: node-to-node distances in periods are precomputed, and at each boundary the joint move minimising the total remaining periods to the drones' goals, subject to connectivity, is chosen, so the measured gap reflects goal selection and not a clumsy execution layer. The implementation is verified by replaying each optimal plan through the same machinery: on all forty instances the replay reproduces the model's optimum exactly, so the optimum is a valid lower bound for every controller in this space. One detail matters for the controllers: in the continuous space a stalled drone is one whose commanded move the projection scales away, and that signal drives both the rule's recovery trigger and the stall feature the learned policy observes; on the lattice a blocked drone simply receives no move, so the same partial-deadlock test, all movers but one held in place, is evaluated once per period and fed to both.

Result. Table 4 and Figure 4 give the gap of each method to the optimum. No run of any method beats the optimum. Over the 47 instances all methods complete, the rule is 55.3% above the optimum and the learned policy 56.0%, indistinguishable ($p = 0.60$); the lookahead rule is 14.3% above it, the visit-order search 23.2%, the task-sequence search 5.9% and the beam search 0.4%, and every one of these is better than the next weaker method at $p < 10^{-4}$ (Wilcoxon). The beam search with $B = 30{,}000$ returns the proven optimum on 39 of the 40 instances without reloading and on 7 of the 10 that force it, in 43 and 121 seconds per instance on average; with $B = 1{,}000$ it is 2% above the optimum without reloading and 23% with it, so the reload instances need the width. The task-sequence search reaches the optimum on 21 of the 40 instances without reloading and is 12–13% above it on the reload instances, partly for want of budget (its gap there falls from 21% to 12% when the budget is raised from 2,400 to 9,000 simulations) and partly because the rule's run, from which it starts, fails to complete on 4 of the 10. The lookahead rule completes all 50, is within 7–15% of the optimum without reloading and 19–29% with it, and returns the optimum on six instances; its cost is 33 seconds per instance without reloading and seven minutes with it.

The methods decompose the gap. Freeing the visit order removes 32 of the rule's 55 points, so the rule's myopic choice of targets is the largest single cost; the lookahead rule, which frees the same choice online by simulation instead of offline by search, removes 41 of the 55, more than the visit-order search because it also chooses the relay mode and the reload moment and re-decides as the state evolves. Freeing the relay decisions, where a relay stands and until when, is worth another 17 points in the task-sequence search, which is what the animated trajectories of the optimal plans suggested: the optimum keeps its relays moving towards where they are needed next, while the rule's relays park on a slot and wait. In the lookahead rule the follow mode captures part of this: with the slot geometry only, the lookahead is 26% above the optimum without reloading, and letting it choose between slot and follow at each decision brings it to 15%. The remaining six points of the task-sequence search and the last fraction of a point of the beam search are the joint timing that only a search over the state itself resolves.

Fine-tuning the learned policy on the small instances. The learned policy was meant to make the target choice and on these instances it does not: it sits with the rule, and on one six-point instance it takes 5,180 steps where the optimum takes 180, which dominates its mean for that row. We therefore continued its training from the proposed weights on the distribution of these instances, three drones, three to eight demand points with capacity 2 above six, random depots, with the validation score measured in the model's decision space so that the selection criterion is the final one, in four runs (two seeds each at regularisation weight 2.0 and 0.5, 600–1,000 episodes to early stopping). All four runs remove the tail, the worst per-instance median falling from 5,180 to at most 600 steps, and none moves the median: the mean gap over the 47 instances is 56.0% before and after ($p = 0.83$), and the per-row medians are unchanged. The policy's option set admits the visit-order search's 23%, so the ceiling is not the interface; within the budget we use, the learner does not find the non-myopic order.

The learned policy with lookahead. Running the rollout lookahead with the proposed learned policy as its base policy, so that the policy proposes the candidate assignments and the rollouts that score them, brings it from 56.0% to 18.9% above the optimum ($p < 10^{-4}$), close to the lookahead rule's 14.3% ($p = 0.011$ in the rule's favour), at 136 seconds per instance without reloading and six minutes with it. The gain comes from the search, not from the learner: with the learned weights as base policy the lookahead is 15.4% above the optimum without

reloading and 36–37% with it, with the rule 12.4% and 19–35%. The learning-plus-search combination is therefore a way to spend compute when it is available, and the quantity it buys is the one the policy alone does not learn.

What the continuous measurement adds. A separate measurement runs the rule and the learned policy in the continuous space they are designed for, against the same lattice optima of the forty instances without reloading. There the rule is 9.7% and the learned policy 4.3% above the optimum, and on the smallest instances they are faster than it. Both statements are true and measure different things. The model's optimum is exact in its own space but is an upper bound on the continuous optimum, because its drones cover h rather than ℓ per period on axis-aligned legs and spend a whole period on each service; so the continuous figures are lower bounds on the true optimality gap, not estimates of it. The large difference between the two columns is not a property of learning: the rule is training-free and loses 48 percentage points in the move to the lattice, against 52 for the learned policy, so all but a few points of the difference are the cost of executing a controller designed for straight-line flight on a 98 m lattice with ten-step periods.

Cost. The searches buy their points with computation: the visit-order search takes a mean of 345 s per instance without reloading and 20 minutes with it, the task-sequence search 298 s and 76 minutes (9,000 simulations), the beam search 43 s and two minutes, the lookahead rule 33 s and seven minutes, against 0.05–0.12 ms per decision for the rule and 2.7–7.4 ms for the learned policy (Table 12); all of them need the instance in advance or a simulator of it. The exact search needs 0–11 s without reloading and 4–26 minutes with it. On these instances the beam search is therefore both the closest to the optimum and cheaper than every other search, because the state space of three drones on fifty nodes is small; its expansion grows as the move set to the power of the fleet size, so it does not extend to the fleets of Section 6.3, whereas the lookahead rule does (Table 2).

Where the reference stops. The exact search is limited by its state space, which grows as 2^M in the demand points and by a factor $(Q + 1)$ per drone once cargo is carried; with $Q = 2$ it handles eight demand points in 22 minutes and we did not push it further. Beyond that we fall back to the MILP on the hull-pruned lattice, one hour per instance and four solver threads, and it stops producing a reference: with eight demand points it returns a feasible solution on 5 of 10 instances and proves optimality on none (mean remaining gap 31%), and with ten it returns one solution out of four (gap 37%). The exact reference therefore stops being available at about six demand points, two orders of magnitude below the instances the controllers are intended to run on.

6.3. RQ3: Comparative performance across operating conditions

Fleet size divides the configuration space. Table 5 and Figure 5 give the paired comparison across the nine configurations. With three drones the rule is faster by 39 and 30 steps in paired median with 20 and 30 demand points. With four drones the two controllers are statistically indistinguishable: on the standard configuration the median paired difference over 480 runs is +2 steps with 234 wins against 239 ($p = 0.52$), and on 4/80 the learned policy is 38 steps faster. With five or more drones the learned policy is faster, by 62 steps at five drones growing to 178 at ten and 76 at twelve (70 wins against 10). The boundary therefore lies at four drones, and the advantage persists beyond the largest fleet we measured, although the monotone growth stops between ten and twelve (Section 6.4 returns to this point). The split concerns the two millisecond controllers; the lookahead rule of Table 2 is faster than both on all three configurations it was run on. An earlier measurement against the rule's pre-tuning recovery parameters gave the same picture (+40, +0, −54, −170 on 3/30, 4/50, 5/50 and 10/300), so the boundary does not depend on the tuning.

Figure 6 shows the full distributions on the three main configurations: with three drones the rule's curve lies to the left, with four the curves coincide, and with five the learned policy's curve lies to the left and reaches completion earlier.

Tails follow the same split. Where drones are scarce the learned policy's worst case is three to four times the rule's (maximum 5,395 versus 1,447 steps on 3/30); where they are plentiful it is lighter (1,101 versus 1,734 on 5/50, 1,365 versus 1,753 on 10/300). On the standard configuration over 480 runs against the pre-tuning rule the learned policy exceeded 1,500 steps five times and the rule fourteen times.

Stabilising the tail. That tail is a liability for a relief operation, where the worst case matters more than the median, so we tried to remove it from both sides. On the *training* side we trained four further runs under the proposed configuration: two with the deadlock-recovery override inside the rollout loop, so that the policy learns against the behaviour it will be deployed with, and two with the entropy target annealed to $0.10 \log n_{\text{valid}}$ instead of 0.25, to make the policy less random late in training. Neither worked. The lower entropy target doubled the spread on 3/30 (standard deviation 629 and 1,001 against 328 for the proposed policy, maxima of 6,283 and 9,049). Of the two runs with recovery in the loop, one matched the proposed policy on 3/30 (+36 against +30) and the other lost 110 steps of median, and evaluating the first of them *without* the override at inference restored the original spread (standard deviation 338, maximum 3,634), which shows that the training change did nothing and the override at inference did the work.

On the *inference* side, applying the rule's recovery step (Procedure 1, step 6) to the proposed policy's options whenever the fleet has been in partial deadlock for k_{stall} steps, with no retraining, removes the tail on every configuration (Table 6). The worst case falls from 4,614 to 1,210 steps on 3/30 and from 3,061 to 1,436 on the standard configuration, the standard deviation falls by a third to a half, and no run exceeds 1,500 steps where twelve did before. The median cost is confined to the configuration where the learned policy was already behind: +64

Table 4: Optimum of the space–time model at h = 98 m versus eight methods, three drones, run in the model's decision space where the optimum is a valid lower bound. Rows with Q = 2 force at least one reload. Each cell is the mean gap over the instances the method completes, with the number of instances on which it returns the proven optimum in parentheses and, where it does not complete all, its completions as a superscript; the last row averages over the instances every method completes. Learned values are medians of 30 runs, the fine-tuned column the median over two seeds of those medians; the visit-order search uses 1,200 simulations, the task-sequence search 2,400 ($M \le 6$) or 9,000 (Q = 2), the beam search B = 30,000.

M	Q	inst.	opt.	mean gap to the optimum (instances on which the method returns the optimum)							
			(mean)	rule	lookahead	order	task seq.	beam	learned	learned, FT	learned+LA
3	5	10	91	+47.5% (0)	+13.4% (1)	+26.8% (0)	+5.1% (6)	+0.0% (10)	+49.5% (0)	+45.8% (0)	+16.2% (1)
4	5	10	115	+45.5% (0)	+6.9% (3)	+15.6% (1)	+2.2% (7)	+0.0% (10)	+46.5% (0)	+44.9% (0)	+8.2% (4)
5	5	10	124	+59.7% (0)	+15.2% (0)	+25.0% (0)	+7.2% (4)	+0.0% (10)	+58.1% (0)	+57.3% (0)	+17.9% (0)
6	5	10	139	+65.5% (0)$^{9/10}$	+14.2% (2)	+21.6% (0)	+5.3% (4)	+0.7% (9)	+333.8% (0)	+68.4% (0)	+19.2% (0)
7	2	5	196	+65.8% (0)$^{4/5}$	+28.7% (0)	+31.2% (0)	+11.9% (0)	+2.1% (3)	+89.8% (0)	+92.0% (0)	+38.3% (0)
8	2	5	234	+55.0% (0)$^{4/5}$	+18.6% (0)	+24.0% (0)	+12.6% (0)	+0.8% (4)	+60.4% (0)	+69.3% (0)	+36.7% (0)
all, every method completes			47	+55.3%	+14.3%	+23.2%	+5.9%	+0.4%	+56.0%	+56.0%	+18.9%

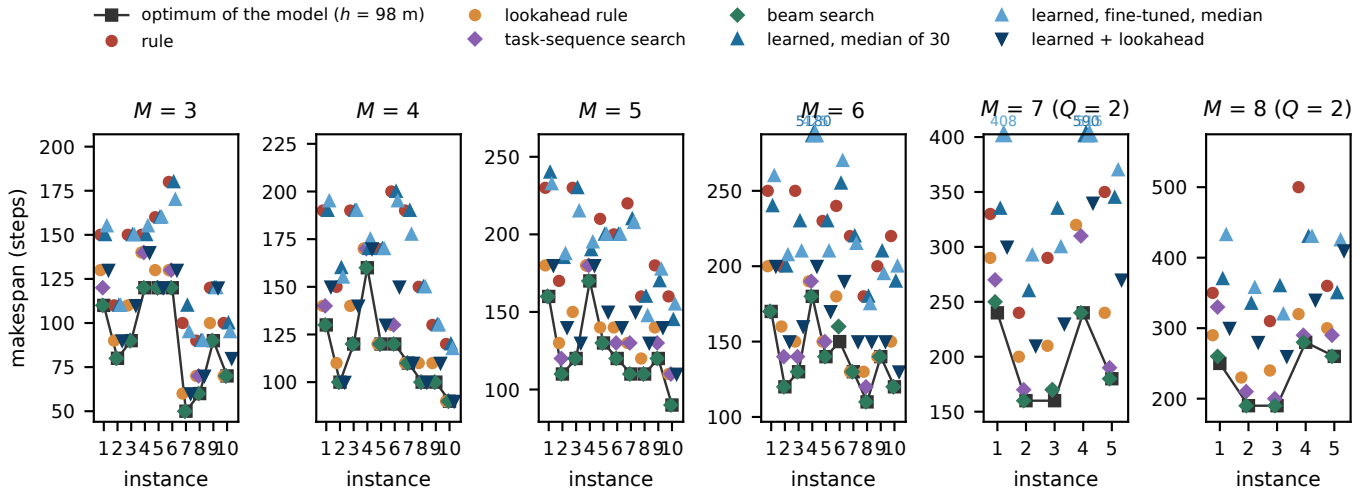


Figure 4: Per-instance makespan on the exactly solved instances with three drones, all methods run in the model's decision space: optimum of the space–time model at h = 98 m (square), rule and lookahead rule (circles), task-sequence and beam search (diamonds), learned policy median of 30 runs, its fine-tuned version and the learned policy with lookahead (triangles); a learned median that leaves the axis is marked at the top edge with its value. Panels with Q = 2 force reloading.

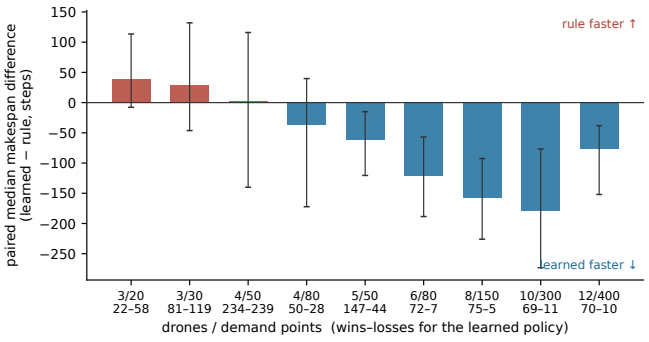


Figure 5: Paired median makespan difference (learned minus rule) by configuration, with the interquartile range of the paired differences as error bars and the learned policy's wins–losses under each label. Negative values favour the learned policy; the boundary lies at four drones.

instead of +30 on 3/30, while the standard configuration improves slightly (−13 against +2, both insignificant) and 5/50 is unchanged (−54 against −62). We therefore report the plain policy as the proposal, because it is the faster one in the median, and recommend the override wherever the worst case is what has to be guaranteed.

Mechanism. The two controllers differ in one decision. On 3/30 the rule stations 1.39 relays on average and the learned policy 0.78; the learned fleet therefore spends 27% of drone-steps stalled by the projection against 8% for the rule. Overriding only the learned policy's relay assignment with the rule's, at inference, recovers the full 44-step gap; matching only the *number* of relays recovers 14 steps of median but removes the tail (maximum from 4,904 to 1,083). The identity of the relaying drone drives the median and the count drives the tail. The information needed to make the rule's choice is present in the learned policy's observation: a two-layer classifier predicts the rule's relay assignment from the 41 observed features with 85% accuracy, and behaviour cloning of the full rule reaches 95% on relay decisions. When drones are plentiful the opposite happens: the rule keeps stationing a chain that the geometry no longer needs, while the learned policy exploits the slack and, with its sampled inference, recoveries the stalls that the deterministic rule repeats; this is also where the rule loses its few completions.

Table 5: Paired comparison on identical instances. Positive differences mean the learned policy is slower than the rule. p from the Wilcoxon signed-rank test on both-completed pairs.

N/M	rollouts	complete h/l	median rule / learned	paired median	win–loss	p
3/20	80	80/80	364 / 420	+39	22–58	$< 10^{-4}$
3/30	200	200/200	572 / 618	+30	81–119	3.4e-04
4/50 (standard)	480	473/480	878 / 878	+2	234–239	0.521
4/80	80	80/80	1,120 / 1,036	−38	50–28	0.006
5/50	200	191/200	588 / 534	−62	147–44	$< 10^{-4}$
6/80	80	79/80	737 / 614	−122	72–7	$< 10^{-4}$
8/150	80	80/80	916 / 722	−158	75–5	$< 10^{-4}$
10/300	80	80/80	1,216 / 1,052	−178	69–11	$< 10^{-4}$
12/400	80	80/80	1,199 / 1,108	−76	70–10	$< 10^{-4}$

Table 6: Effect of the deadlock-recovery override applied to the proposed policy at inference, without retraining. The paired median, wins–losses and p are against the tuned rule on the same instances; the standard deviation, 90th percentile, maximum and the count above 1,500 steps describe the learned policy's own makespan distribution.

config.	inference	runs	paired median	wins–losses	p	s.d.	90th pct.	max	>1,500
3/30	proposed	200	+30	81–119	3.4e-04	328	837	4614	2
	+ recovery	200	+64	66–134	$< 10^{-4}$	149	858	1210	0
4/50	proposed	480	+2	234–239	0.521	208	1104	3061	7
	+ recovery	240	−13	125–110	0.249	132	1052	1436	0
5/50	proposed	200	−62	147–44	$< 10^{-4}$	125	712	1302	0
	+ recovery	200	−54	145–44	$< 10^{-4}$	131	734	1075	0

Why the relay-limited deficit resists policy-side fixes. Table 7 lists eleven interventions, each trained with four or two seeds under the same training configuration except for the stated change and evaluated with the same paired protocol, that attempt to make the learned policy relay like the rule when drones are scarce. The variants are, in the numbering of the table: (V1) the cross-entropy regularisation towards the rule is weighted three times higher for drones whose rule option is relay; (V2) each drone on the communication path of a delivering teammate receives an individual reward (2.0 or 6.0) at that delivery and the critic is trained on per-drone returns; (V3) checkpoint selection scores half of its validation instances on the 3/30 configuration instead of only on 4/50, eight seeds; (V4) as V2 but zero-sum, the delivering drone losing what its relays gain; (V5) the critic is conditioned on the teammates' chosen roles (a joint-action critic) with the teammates' next roles sampled from the actor in the target; (V6) the autoregressive order is reversed to decide the drone nearest the depot first; (V7) all goals are invalidated after 60 consecutive partial-deadlock steps so that the manager re-decides, in training and evaluation; (V8) the mean over per-drone action values in the critic is replaced by state-dependent monotone weights that sum to one with a floor of one half; (V9) each drone's option is stored as its own single-agent transition, chained inside a decision point with discount one, so that no decomposition of the team return is needed; (V10) the manager is initialised by behaviour cloning of the rule on 40,000 decisions (92.5% accuracy, 95% on relay decisions) and then fine-tuned, with and without a 50-episode critic warm-up; (V11) a per-step penalty proportional to the estimated arrival time of the bottleneck demand point, with the estimate divided by each drone's realised-movement ratio so that blocked drones count as slow. None improves the 3/30 median and several hurt the configurations where the learned policy wins. The pattern is informative. Adding reward for relaying turns the means into the end: the policy relays for its own sake and 5/50 reverses from −54 to +60. Every route to per-drone credit is blocked for a different reason: a joint-action critic inherits the variance of a sampled teammate action in its target; a monotonic mixing network either diverges when its weights are free or collapses to the mean when they are floored; a sequential single-agent reformulation removes the decomposition but leaves an undiscounted credit chain with no contraction. Initialising the policy by behaviour cloning of the rule, with 95% relay accuracy, is erased within 400 episodes: once the actor is updated, relay usage falls to the level of a randomly initialised policy within 50 episodes and stays there. A penalty on the bottleneck demand point's estimated arrival makes the policy reduce the *average* rather than the maximum. We conclude that, within this policy structure and objective, under-relaying is the learner's fixed point, not an artefact of credit, information or initialisation.

Sensitivity to the connectivity enforcement model. The projection of Section 4.1 scales a blocked displacement along its own direction, so a drone pulling on the chain effectively stops. We implemented an alternative that clamps the target position to the nearest point of the communication disc around the binding neighbour, which removes only the outward component and keeps the tangential one (a drone can still circle at constant range). Both controllers use the same projection in each comparison, and connectivity is still never violated. These measurements predate the rule tuning of Section 6.5 and use $k_{\text{stall}} = 60$,

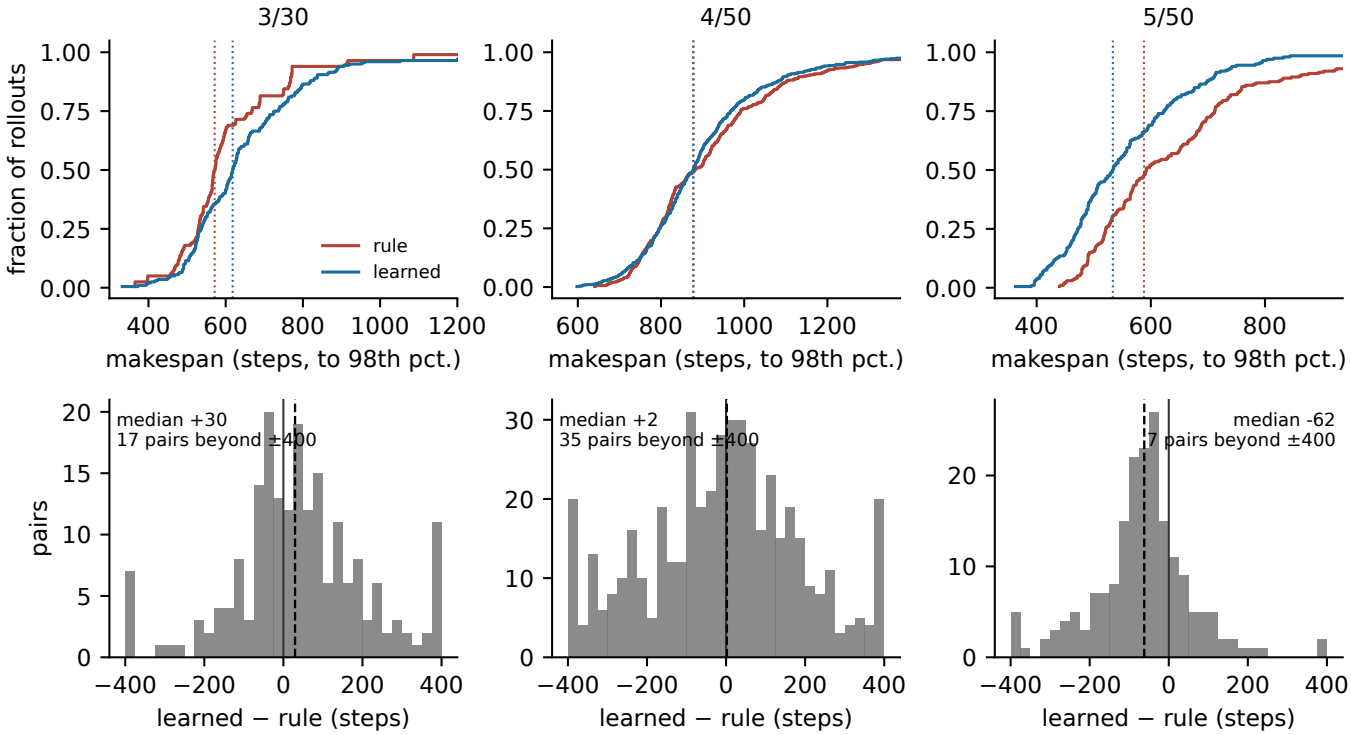


Figure 6: Top: cumulative distribution of makespan on 3/30, 4/50 and 5/50 (dotted lines mark medians; axes cut at the 98th percentile). Bottom: distribution of the paired difference on both-completed instances, clipped to ±400 steps with the number of clipped pairs noted.

$h_{\text{hold}} = 40$ for both projections, so they are internally consistent. Evaluating the same learned weights under the sliding projection without retraining reduces the 3/30 deficit from +40 to +21; retraining four seeds under it gives +15 to +48 (mean +28) and lowers the learned fleet's stall ratio from 0.27 to 0.17–0.22, with completion still 100%. The rule gains little with three drones (median 570 to 571) but gains more than the learned policy with four (866 to 812 against 879 to 850), so the standard-configuration equality becomes +26 in its favour. We therefore keep the radial projection as the primary model and report this as a sensitivity result: about half of the three-drone deficit is a property of how the constraint is enforced rather than of the learner.

Communication radius. Both controllers were run without retraining at $R = 250$ m and $R = 350$ m (Table 8, Figure 7); the learned policy normalises distances by R, and the instance generator rescales the reachable region $0.9NR$ with R, so both controllers see the same instances at each radius. Completion transfers: the learned policy completes every instance at both radii, while the rule loses 18 of 200 instances on 5/50 at the shorter radius. Speed shows two effects. The first is the regime shift seen with fleet size: at $R = 250$ m a demand point needs more hops, the fleet is effectively scarcer, and every configuration moves towards the rule (+77 on 3/30, +68 on 4/50, and the 5/50 advantage shrinks to an insignificant −16 ($p = 0.33$)); at $R = 350$ m the fleet is effectively more plentiful and the learned policy is faster already with four drones (−40 on 4/50). The second is a training-radius effect: on 3/30 and 5/50 the learned policy's margin is best at the radius it was trained on and worse on both sides (+30 at 300 m against +77 and +52; −62 against −16 and −42), so the normalisation by R transfers completion but not the full speed. The regime variable is therefore the chain demand, the hops a demand point needs relative to the drones available, rather than the fleet size alone, and a deployment at a radius far from the training radius should retrain or at least re-evaluate the boundary.

6.4. RQ4: Upward transfer of the learned policy

The learned policy never saw more than four drones or more than 80 demand points in training, yet it completes every instance with 5 to 12 drones and 50 to 400 demand points and is faster than the rule on all of them. Two properties carry this. The set-based encoder has no parameter that depends on N or M, so larger fleets are simply larger sets; and training samples the fleet size and depot afresh every episode. Table 9 separates the two by running the ablation policy that was trained on the fixed 4/50 configuration alone (no configuration randomisation) on the same large-fleet instances without retraining. That policy also completes every instance from 5 to 10 drones, so completion generalises through the architecture alone; but it is slower than the randomised policy on every configuration (no significant difference from the rule at 5/50 against −62, and −51 to −174 against −122 to −178 beyond), so speed on large fleets is learned from the variety of small ones. The same ablation loses 114–296 steps on 3/30 (Table 11), so randomisa-

Table 7: Interventions that did not remove the relay-limited deficit, numbered as in the text. Cells give the range over seeds of the paired median difference against the rule (positive: learned slower) and, in parentheses, the number of seeds whose difference is significant at $p < 0.05$ (Wilcoxon signed-rank) over the number of seeds trained. The reference row is the proposed policy against the tuned rule (Table 5); the variant rows were evaluated against the rule with its pre-tuning recovery parameters ($k_{\text{stall}} = 60$, $h_{\text{hold}} = 40$), against which the proposed policy measured +40, −54 and +0, so differences of a few tens of steps between the reference row and a variant row are not meaningful.

#	Intervention (what changed)	3/30	5/50	4/50	Mechanism of failure
—	Proposed configuration (reference)	+30 (1/1)	−62 (1/1)	+0 (0/1)	—
V1	Relay-weighted rule regularisation (×3)	+18, +58 (1/2)	−29, +13 (0/2)	+13, +41 (1/2)	relay-specific regularisation harms
V2	Per-drone relay credit (2.0 / 6.0)	+36 to +82 (4/4)	+60 to +106 (4/4)	+39 to +131 (4/4)	relaying becomes the goal
V3	Checkpoint selection half on 3/30	+37 to +105 (8/8)	−58 to −22 (7/8)	−37 to +108 (3/8)	improves only the standard config.
V4	Zero-sum credit redistribution	+91, +126 (2/2)	+98, +172 (2/2)	+146, +180 (2/2)	same, larger
V5	Joint-action critic	+71 to +124 (4/4)	−46 to −16 (3/4)	+8 to +100 (2/4)	target variance from sampled teammates
V6	Nearest-first decision order	+46, +100 (2/2)	−43, −38 (2/2)	+4, +106 (1/2)	order is not the cause
V7	Re-decision after 60 stalled steps	+82, +95 (2/2)	−38, −34 (2/2)	+17, +55 (0/2)	irregular option lengths
V8	Monotone mixing critic (floor 0.5)	+54 to +230 (4/4)	−28 to +38 (2/4)	+55 to +258 (4/4)	divergence vs. credit on one axis
V9	Sequential single-agent credit	+81 to +213 (4/4)	+8 to +27 (2/4)	+62 to +170 (4/4)	no contraction within a decision
V10	Behaviour-cloning initialisation (± warm-up)	+6 to +69 (3/4)	−42 to −20 (3/4)	−16 to +40 (0/4)	erased within 400 episodes
V11	Bottleneck-arrival penalty (0.15 / 0.25)	+85 to +144 (4/4)	−32 to +13 (2/4)	+7 to +156 (3/4)	reduces the mean, not the maximum

Table 8: Transfer to other communication radii without retraining. Cells give completions (rule/learned) and the paired median difference with win–loss and p; $R = 300$ m is the training radius.

N/M	$R = 250$ m	$R = 300$ m	$R = 350$ m
3/30	200/200 / +77 (45–154, $< 10^{-4}$)	200/200 / +30 (81–119, 3.4e-04)	200/200 / +52 (58–141, $< 10^{-4}$)
4/50	118/120 / +68 (40–77, $< 10^{-4}$)	473/480 / +2 (234–239, 0.521)	119/120 / −40 (77–42, 1.1e-04)
5/50	182/200 / −16 (99–83, 0.327)	191/200 / −62 (147–44, $< 10^{-4}$)	200/200 / −42 (152–46, $< 10^{-4}$)

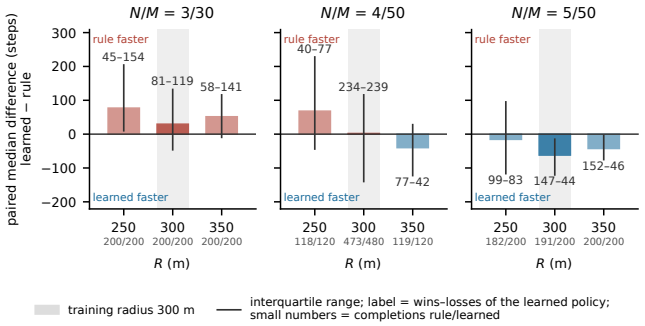


Figure 7: Transfer to other communication radii without retraining, one panel per configuration. Bars are the paired median difference (learned − rule, steps): a bar above zero (red) means the rule was faster, below zero (blue) the learned policy; whiskers are the interquartile range of the paired differences, the label at each bar is the learned policy's wins–losses, and the completions (rule/learned) are printed under the radius. The shaded column is the training radius. Reading left to right within a panel: a shorter radius moves every configuration towards the rule, and on 3/30 and 5/50 the learned policy's margin is best at the radius it was trained on.

tion is what makes the small-fleet side usable at all. Both policies were trained with at most four drones; the hop-depth and slot features are normalised by a constant of eight, so fleets of ten and twelve drones also take the encoder outside its training range of feature values, and the policy still completes them.

Where the transfer weakens. We have not isolated the cause of the narrowing at 12/400 and offer three conjectures. First, both controllers maintain a single chain to one lead, so with twelve drones most of the fleet is delivering near the depot regardless of the controller and the decisions that differ between them concern a shrinking share of the work. Second, with 400 demand points a drone's five candidates are all close to it, so the choice among them matters less and the learned policy's advantage, which lies in choosing targets and relays jointly, has less to act on. Third, the manager's hop-depth and slot features are normalised by a constant of eight and exceed one for larger fleets, and twelve sequential autoregressive decisions compound sampling noise; both take the policy farther from its training distribution. Which of these dominates would require fleets beyond twelve drones to settle.

6.5. RQ5: Are the components necessary?

Rule. Table 10 grids $k_{\text{stall}} \in \{30, 60, 90, 120\}$, $h_{\text{hold}} \in \{20, 40, 80\}$ and the two recovery modes, plus no recovery, on 3/30, 4/50 and 5/50 with 40 tuning seeds each, and ranks settings by total completions first and the sum of median makespans second; this criterion was fixed before the measurement. The recovery mechanism is necessary: without it the rule completes 25–26 of 40 standard instances (two measurements). The mode matters more than the timing: every lead-swap setting completes 117–120 of 120 instances, while retreat settings complete 101–117 because slackening the chain does not change the assignment that caused the stall. Within the lead-swap family the differences are not statistically meaningful: in a second measurement of the grid, reported in the table together with a paired Wilcoxon test of every setting against the selected one on the same 40 instances, no setting differs from the selected one at $p < 0.05$ on any configuration (all $p \geq 0.09$), and one other setting (30/40) also completes all 120 with a median sum 13 steps lower, which is tie-breaking noise. The selected setting ($k_{\text{stall}} = 120$, $h_{\text{hold}} = 20$) was fixed from the first measurement and is used throughout. The adaptive relay count is necessary as well: fixing one or two relays completes 0–65% (Table 3).

Table 9: Transfer to fleets larger than any seen in training, without retraining. Cells give the learned policy's completions, the paired median difference against the rule (negative: learned faster), and in parentheses the learned policy's wins–losses and the Wilcoxon signed-rank p. The proposed policy is trained on $N \in \{3, 4\}$ with random depots; the comparison policy is the ablation trained on the fixed 4/50 configuration only (two seeds).

N/M	trained on $N \in \{3,4\}$, $M \in [20, 80]$	trained on 4/50 only, seed 1	seed 2
5/50	200/200 / −62 (147–44, $< 10^{-4}$)	80/80 / +7 (35–39, 0.840)	80/80 / +44 (28–50, 0.002)
6/80	80/80 / −122 (72–7, $< 10^{-4}$)	80/80 / −56 (57–20, $< 10^{-4}$)	80/80 / −51 (51–27, 0.010)
8/150	80/80 / −158 (75–5, $< 10^{-4}$)	80/80 / −129 (68–11, $< 10^{-4}$)	80/80 / −110 (60–20, $< 10^{-4}$)
10/300	80/80 / −178 (69–11, $< 10^{-4}$)	80/80 / −152 (69–11, $< 10^{-4}$)	80/80 / −174 (68–12, $< 10^{-4}$)

Table 10: Rule ablation on tuning seeds disjoint from the evaluation seeds (40 instances per configuration). Cells give completed instances, the median makespan over completed instances and, in parentheses, the p-value of a Wilcoxon signed-rank test of that setting against the selected setting (bold) on the instances both completed; the last column sums completions over the three configurations.

mode	k_{stall} / h_{hold}	3/30	4/50	5/50	completions
none	—	39/40 / 559 (0.228)	25/40 / 817 (0.258)	38/40 / 602 (0.093)	102/120
lead swap	30 / 20	40/40 / 557 (0.121)	39/40 / 853 (0.130)	40/40 / 625 (0.928)	119/120
lead swap	30 / 40	40/40 / 562 (0.673)	40/40 / 867 (0.775)	40/40 / 606 (0.247)	120/120
lead swap	30 / 80	40/40 / 567 (0.158)	39/40 / 844 (0.754)	39/40 / 614 (0.404)	118/120
lead swap	60 / 20	40/40 / 557 (0.599)	39/40 / 849 (0.780)	40/40 / 600 (0.791)	119/120
lead swap	60 / 40	40/40 / 554 (0.497)	39/40 / 852 (0.807)	40/40 / 625 (0.944)	119/120
lead swap	60 / 80	40/40 / 556 (0.359)	39/40 / 863 (0.788)	39/40 / 632 (0.675)	118/120
lead swap	90 / 20	40/40 / 566 (0.206)	39/40 / 871 (0.908)	40/40 / 599 (0.375)	119/120
lead swap	90 / 40	40/40 / 559 (0.473)	40/40 / 856 (0.674)	37/40 / 634 (0.844)	117/120
lead swap	90 / 80	40/40 / 563 (0.786)	39/40 / 858 (0.149)	39/40 / 629 (0.615)	118/120
lead swap	**120 / 20**	40/40 / 561	40/40 / 867	40/40 / 620	**120/120**
lead swap	120 / 40	40/40 / 557 (0.757)	39/40 / 878 (0.685)	40/40 / 624 (0.979)	119/120
lead swap	120 / 80	40/40 / 562 (0.687)	38/40 / 829 (0.546)	39/40 / 626 (0.850)	117/120
retreat	30 / 20	37/40 / 559 (0.350)	35/40 / 836 (0.185)	40/40 / 632 (0.528)	112/120
retreat	30 / 40	39/40 / 556 (0.079)	29/40 / 832 (0.284)	39/40 / 621 (0.157)	107/120
retreat	30 / 80	38/40 / 561 (0.754)	26/40 / 804 (0.443)	37/40 / 584 (0.368)	101/120
retreat	60 / 20	39/40 / 556 (0.866)	38/40 / 847 (0.215)	38/40 / 603 (0.759)	115/120
retreat	60 / 40	39/40 / 559 (0.875)	39/40 / 823 (0.485)	39/40 / 624 (0.459)	117/120
retreat	60 / 80	39/40 / 563 (0.991)	26/40 / 795 (0.236)	40/40 / 635 (0.490)	105/120
retreat	90 / 20	39/40 / 566 (0.512)	38/40 / 822 (0.058)	40/40 / 642 (0.773)	117/120
retreat	90 / 40	39/40 / 566 (0.180)	39/40 / 852 (0.634)	39/40 / 618 (0.615)	117/120
retreat	90 / 80	39/40 / 562 (0.665)	36/40 / 830 (0.578)	37/40 / 598 (0.354)	112/120
retreat	120 / 20	39/40 / 556 (0.645)	37/40 / 863 (0.327)	38/40 / 614 (0.460)	114/120
retreat	120 / 40	39/40 / 560 (0.757)	37/40 / 831 (0.451)	38/40 / 628 (0.602)	114/120
retreat	120 / 80	39/40 / 561 (0.191)	38/40 / 854 (0.815)	37/40 / 627 (0.308)	114/120

Learned policy. Table 11 removes one element at a time, retrains two seeds with the same budget, and runs each ablated policy on the same instances as the proposed policy (the first 120, 80 and 80 paired runs of Section 6.3); cells give the median of the per-instance makespan difference and a Wilcoxon signed-rank p, per seed. Rule regularisation and entropy-target annealing are necessary everywhere: removing either costs 52–189 steps on every configuration and seed at $p < 10^{-4}$. Configuration randomisation is necessary off the training distribution: removing it is neutral on the standard configuration (−19 and −34, $p \geq 0.12$), on which the ablated policy was trained, but costs 114–296 steps on 3/30 and 53–76 on 5/50. Autoregressive decisions help on every configuration by 32–88 steps ($p \leq 0.007$). The residual penalty is neutral: no seed on any configuration differs from the proposed policy at $p < 0.05$. A manager with a fixed-size MLP encoder instead of the set encoder, trained in an earlier version on the standard configuration only, completed 38% of standard instances and could not be run on other fleet sizes at all.

6.6. Inference cost

Table 12 gives the latency of one manager decision on a CPU. The rule costs 0.05–0.12 ms; the learned policy 1.3–2.5 ms without autoregression and 2.7–7.4 ms with it, growing linearly with the fleet because each drone is a sequential network call. Over a 900-step episode the learned policy's cumulative decision time is 0.16 s. The lookahead rule spends 59, 302 and 361 seconds per instance on 3/30, 4/50 and 5/50 (Table 2), that is, 0.1–0.5 seconds of simulation per decision point. The exact search takes 0–11 seconds on instances with three to six demand points at $h = 98$ m, and the MILP for larger instances uses its full hour.

Table 11: Ablation of the learned policy. Each cell gives, for two independently trained seeds, the median per-instance makespan difference between the ablated policy and the proposed policy on the same instances (positive: the removed element was needed) and the Wilcoxon signed-rank p.

Removed component	4/50 (120 runs)	3/30 (80)	5/50 (80)
Rule regularisation	+121 ($< 10^{-4}$) / +189 ($< 10^{-4}$)	+171 ($< 10^{-4}$) / +160 ($< 10^{-4}$)	+52 ($< 10^{-4}$) / +152 ($< 10^{-4}$)
Configuration randomisation	−19 (0.260) / −34 (0.119)	+114 ($< 10^{-4}$) / +296 ($< 10^{-4}$)	+53 (2.7e-04) / +76 ($< 10^{-4}$)
Entropy target annealing	+136 ($< 10^{-4}$) / +138 ($< 10^{-4}$)	+138 ($< 10^{-4}$) / +130 ($< 10^{-4}$)	+58 ($< 10^{-4}$) / +76 ($< 10^{-4}$)
Autoregressive decisions	+32 (0.007) / +84 ($< 10^{-4}$)	+88 (3.2e-04) / +56 (0.002)	+42 (2.7e-04) / +60 (9.1e-04)
Residual penalty	−4 (0.746) / −32 (0.204)	+16 (0.625) / −28 (0.291)	+24 (0.120) / +12 (0.374)

Table 12: Latency of one manager decision (milliseconds, CPU) and the cumulative decision time of a 900-step episode.

Controller	$N = 3$	$N = 4$	$N = 6$	$N = 8$	900 steps
Geometric rule + recovery	0.05	0.08	0.10	0.12	0.003 s
Learned, independent decisions	1.27	1.52	1.94	2.45	0.069 s
Learned, autoregressive (proposed)	2.70	3.52	5.32	7.35	0.158 s

7. Discussion

Deployment. If a few minutes of simulation per instance are affordable, the lookahead rule is the controller to run: it completes everything we gave it and is 13–30% faster than either millisecond controller on every configuration measured. When decisions must be made in milliseconds, the two online controllers are complementary and the boundary is sharp, so a fleet operator can choose by fleet size: the rule with three drones, the learned policy with five or more, where it is both faster and more reliable; with four the two are statistically indistinguishable in speed and the learned policy completes more. All of them keep the fleet connected by construction.

What the negative results mean. Eleven interventions that change how the learner is credited, informed, initialised or rewarded leave the relay-limited deficit in place, and a change to the environment's enforcement model removes half of it. The deficit is therefore best understood as an interaction between the learner's fixed point under this objective and the geometry of the constraint, not as a shortcoming that more training or a better critic would fix. The same reading applies to the gap to the optimum: fine-tuning on the exact instances removes the policy's tail and leaves its median where the rule is, while a one-step lookahead removes most of the gap from either base policy, so what the learner lacks is the non-myopic target choice that simulation supplies. For a practitioner the actionable consequence is the one above: pick the controller by the latency budget first and by regime second.

Limitations. The learned policy was trained at a single communication radius; it transfers to radii of 250 and 350 m in completion but its speed ranking against the rule shifts with the radius (Section 6.3), and radii outside this range were not tested. The environment is planar, ignores wind, battery and collisions, and uses a disc communication model; real links are not discs. The exact optimum is that of a discretised model at a single lattice spacing: the 98 m lattice and the ten-step period leave a quantisation that refining to 49 m shrinks by 7–9% but that we cannot afford on every instance, so the measured gaps understate the true ones, and beyond eight demand points only solver incumbents with a remaining optimality gap are available. The beam search inherits the exact solver's state representation and does not extend to large fleets; the lookahead rule does, but at a cost that grows with the episode length and the fleet. The rule's novelty lies in the recovery mechanism and the adaptive relay count rather than in the chain geometry itself.

Methodological notes. Three practices mattered for the conclusions. Completed-episode averages favour the controller that gives up, so comparisons must be paired. Forty runs cannot support claims about tails; several of our own interim conclusions reversed when the sample grew. And a pre-check that uses a different metric from the final judgement will pass interventions that the final judgement rejects.

8. Conclusion

We proposed two controllers for connectivity-constrained drone delivery, a geometric chain-relay rule with deadlock recovery and a fleet-size-agnostic learned policy, both acting through a projection that keeps the fleet connected at every step. Under a paired protocol across fleets of three to twelve drones the learned policy completes every instance and is faster by 38 to 178 steps once more than four drones are available, although it was trained on three and four, while the rule is faster by 30–39 steps with three. The relay-limited deficit is one decision, which drone relays; it resists eleven policy-side interventions and halves when the enforcement model is changed. Where an exact optimum of the space–time model is available, both online controllers are about 55% above it in the model's own decision space; a one-step rollout lookahead on the rule is 14% above it, offline searches over visit orders and over task sequences 23% and 6%, and a beam search in the model's state space 0.4%, returning the proven optimum on 46 of 50 instances. The headroom lies first in choosing targets and then in moving relays, and simulation supplies the first part to either base policy: the lookahead rule is also 13–30% faster than both online controllers in the continuous space, at minutes per instance, and the learned policy with the same lookahead is 19%

above the optimum, whereas fine-tuning the policy on the exact instances removes its tail and not its median. The case for the learned policy is therefore its millisecond latency and its transfer to large fleets, where neither the exact search nor the beam search reaches. The natural next step is a policy whose output is a team assignment rather than a set of per-drone options, so that the relay identity becomes a coordinate the critic can value directly, trained with the lookahead's choices as its target.

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[To be added.]

Data and code availability

Code, trained weights, instance generators and the paired-comparison records are available at `https://github.com/koptimizer/relay_drone_network`.

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